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Invention, Authorship, "Intellectual Property," and the Origin of Patents: Notes toward a Conceptual History

PAMELA O. LONG

Intellectual property issues involving both old and new technologies are pressing and controversial in contemporary scientific and technical culture.[1] Yet the historical development of the concept of intellectual property itself needs far more study. As it pertains to technology, "intellectual property" involves particular attitudes toward craft knowledge and practice, invention, and authorship, and is properly studied within the context of these intrinsically related issues. A complex background of intellectual history precedes the origin of the limited monopoly that is called a patent.

"Intellectual property," a legal concept, refers to various kinds of *intangible* property. With reference to technology, it involves the belief that knowledge of craft processes and techniques and the development of technological innovations are forms of property with commercial value that are separate from products or devices.[2] Intellectual property laws grant limited monopolies under certain conditions for particular kinds of authorship. Most modern legal systems distinguish between patents and copyrights. Yet to project this distinction too

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[1]For an introduction to recent issues and controversies with extensive further bibliography, see Dorothy Nelkin, *Science as Intellectual Property: Who Controls Research?* (New York, 1984); *Science, Technology, and Human Values*, vol. 10 (Spring 1985), an issue devoted to openness and secrecy in science and technology; and Vivian Weil and John W. Snapper, eds., *Owning Scientific and Technical Information: Value and Ethical Issues* (New Brunswick, N.J., 1989).

[2]A useful discussion of conceptual issues is Rochelle Cooper Dreyfuss, "General Overview of the Intellectual Property System," in Weil and Snapper, eds., pp. 17–40. Because intellectual property is a form of property, the idea of private property itself is germane. For an introduction, see Richard Schlatter, *Private Property: The History of an Idea* (1951; reprint, New York, 1973).

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sharply into the historical past is anachronistic. Authorship can refer to written authorship or the authorship of making and inventing material things. Historically, patents appeared before copyrights, but the concept of "intellectual property" (a kind of intangible property) emerged prior to either form of limited monopoly. Novelty (that which is new) preceded originality (that which originates in a particular author) as the justification for patent awards.[3] Yet the connection between individual authorship and intellectual property had been made by the 15th century at the latest, first with regard to material inventions and later (in the 16th century) with regard to writings.[4]

[3]The distinction between novelty and originality, crucial for modern intellectual property law, is also useful for historical investigations. For instance, a U.S. patent must meet tests for both originality and novelty. The invention must originate with the inventor (it must be original). Yet it must also be novel, i.e., it cannot have been invented independently by someone else. On the other hand, a U.S. copyright must be original but does not have to be novel. In the unlikely event that I wrote a play called *Hamlet* exactly like the one written by Shakespeare without any knowledge or influence of the prior work, I would own the copyright in that work as my own original (but not novel) expression. This is why the U.S. Patent Office (presumably) does not have a file on people who claim to have invented the wheel, whereas the U.S. Copyright Office does have a file on people who insist that they have written an "original" song called "Happy Birthday to You." For further discussion of the distinction, see Arthur Kuflik, "Moral Foundations of Intellectual Property Rights," in Weil and Snapper, eds., pp. 219–40, esp. 236–39.

[4]Thus author's copyright—that is, the grant of a limited monopoly to writings on the basis of authorship—had its genesis in the 16th century. Early copyrights were often granted to printers, but also to authors. Here I am in disagreement with recent claims made on theoretical grounds for the 18th-century origin of author's copyright by scholars who have been influenced by an analysis of authorship by Michel Foucault, "What Is an Author?" in *Language, Counter-Memory, Practice: Selected Essays and Interviews*, ed. Donald F. Bouchard, trans. Donald F. Bouchard and Sherry Simon (Ithaca, N.Y., 1977). See esp. Carla Hesse, "Enlightenment Epistemology and the Laws of Authorship in Revolutionary France, 1777–1793," *Representations* 30 (Spring 1990): 109–37; Mark Rose, "The Author as Proprietor: *Donaldson v. Becket* and the Genealogy of Modern Authorship," *Representations* 23 (Summer 1988): 51–85; and Martha Woodmansee, "The Genius and the Copyright: Economic and Legal Conditions of the Emergence of the Author," *Eighteenth-Century Studies* 17 (Summer 1984): 425–48. The work of these scholars constitutes significant and sophisticated analysis of 18th-century copyright issues. In my view, however, they oversimplify the prior history of intellectual property and authorship. They are able to do this by defining copyright too narrowly as the moral right of authors over their own literary productions within the framework of Lockean notions of natural rights, and the 18th-century commercial book market. Yet copyright concerns the notions of intangible property, limited monopoly, social utility, and the prerogatives of authors over their writings before John Locke's formulation of natural rights, as well as the issues noted above. Even for the 18th century, as the legal historian Jane C. Ginsburg has demonstrated, author's rights do not always occupy a central place within copyright legal history ("A Tale of Two Copyrights: Literary Property in Revolutionary France and America," *Tulane Law Review* 64 [May 1990]:

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The concept of intellectual property has not existed at all times or in all places. Our understanding of its historical development stands at a relatively rudimentary stage. Because intellectual property attitudes constitute a particular approach to craft knowledge, authorship, and invention, beliefs about these larger issues are the initial focus of this article. I suggest here that although some of the components of the notion of "intellectual property" are evident in antiquity, the fully developed concept first emerges in the medieval period around the 12th or 13th centuries.

Positive Traditions concerning the Origin and Status of the Practical Arts in Greek and Roman Antiquity

The positive valuation of craft knowledge and material invention is a basic precondition for the development of intellectual property attitudes with regard to them. Although their intrinsic value may seem obvious, some aristocratic traditions in the ancient world encouraged contempt for handwork and practice. Manual work was associated with slave labor. Moreover, interest in the phenomenal and material world was scorned by those philosophical traditions, such as the Eleatic, which proposed that reality rested in the unseen or in ideal forms. Until recently, however, historians have tended to overgeneralize concerning ancient traditions of contempt for handwork. They have minimized or overlooked the positive ancient traditions of technology and practice.[5]

991–1031). Author's copyright developed after the invention of the printing press, initially in the 16th century. Its emergence was influenced by the commercial book market that resulted from the invention of printing. The issues discussed in this article on patents were also germane to that development.

[5]Elspeth Whitney, *Paradise Restored: The Mechanical Arts from Antiquity through the Thirteenth Century* (Philadelphia, 1990), esp. pp. 23–55, has recently discussed this exaggeration of ancient contempt in a study of the classification of the arts in antiquity and the Middle Ages. For the low status of handwork in antiquity, see Alison Burford, *Craftsmen in Greek and Roman Society* (Ithaca, N.Y., 1972); Arthur T. Geoghegan, *The Attitude towards Labor in Early Christianity and Ancient Culture* (Washington, D.C., 1945); and Paolo Rossi, *Philosophy, Technology, and the Arts in the Early Modern Era,* trans. Salvator Attanasio, ed. Benjamin Nelson (New York, 1970), pp. 13–15. Ancient views toward work, practice, and technique are relevant. The studies of Rodolfo Mondolfo are fundamental—see esp. the useful collection of articles and introduction to his work in Rodolfo Mondolfo, *Polis, lavoro e tecnica,* ed. Massimo Venturi Ferriolo (Milan, 1982). See also the still-important review essay of Ludwig Edelstein, "Recent Trends in the Interpretation of Ancient Science," in *Roots of Scientific Thought: A Cultural Perspective,* ed. Philip P. Wiener and Aaron Noland (New York, 1957), pp. 90–121; and, in addition, Peter Caws, "Praxis and Techne," in *The History and Philosophy of Technology,* ed. George Bugliarello and Dean B. Doner (Urbana, Ill., 1979), pp. 227–37; Moses I.

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One such positive tradition emphasized that progress was indebted to the gradual *human* invention and discovery of the various crafts. This view suggested not only that progress had occurred but that it was brought about by human ingenuity rather than divine beneficence. The first evidence that we have for such an outlook exists in a fragment by the late-6th/early-5th-century Ionian poet/philosopher Xenophanes. An exile who lived in Sicily, Xenophanes wrote: "Not all things, by any means, did the gods show to mortals: rather, as time went on, men found improvement by constant searching."[6] Xenophanes undoubtedly met Athenians in Sicily. Whether or not he directly influenced some of them, views similar to his were well established in Athens by the mid-5th century B.C.

It was at this time that Pericles (ca. 495–429 B.C.) initiated the great public building program that produced the Parthenon and other magnificent monuments and sculptures. The Periclean program was based on and surrounded by numerous well-developed artisan crafts, such as stonecutting, carpentry, pottery, and painting. It points to a high degree of civic confidence and a deep appreciation for the

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Finley, "Technical Innovation and Economic Progress in the Ancient World," *Economic History Review,* 2d ser. 18 (1965): 29–45; Desmond Lee, "Science, Philosophy, and Technology in the Greco-Roman World," *Greece and Rome* 20 (April 1973): 65–78, and 20 (October 1973): 180–93; Reimar Müller, *Poiesis-Praxis-Theoria: Zur Bewertung der Technik in der Kulturtheorie der griechischen Antike* (Berlin, 1989); and Wolfgang Schadewaldt, "The Concepts of Nature and Technique according to the Greeks," in *Research in Philosophy and Technology,* ed. Paul T. Durbin and Carl Mitcham (Greenwich, Conn., 1979), 2:159–71.

[6]See D. J. Conacher, *Aeschylus' "Prometheus Bound": A Literary Commentary* (Toronto, 1980), p. 84, who also provides a useful discussion of the subsequent development of Xenophanes' ideas in the 5th century B.C. tragedy. I have used Conacher's translation of the fragment, which is published in Hermann Diels and Walther Kranz, eds., *Die Fragmente der Vorsokratiker,* 3 vols., 10th ed. (Berlin, 1961), 1:113–39, for Xenophanes' life and extant fragments, and frag. B18 for the citation. For further discussion of Xenophanes and the tradition of belief in human progress that he seems to have influenced, see *Encyclopedia of Philosophy,* s.v. "Xenophanes of Colophon," by G. B. Kerferd; and E. R. Dodds, *The Ancient Concept of Progress and Other Essays on Greek Literature and Belief* (Oxford, 1973), pp. 1–44; Ludwig Edelstein, *The Idea of Progress in Classical Antiquity* (Baltimore, Md., 1967), pp. 3–19; William K. C. Guthrie, *In the Beginning: Some Greek Views on the Origins of Life and the Early State of Man* (1957; reprint, Westport, Conn., 1986), pp. 80–94; G. S. Kirk, J. E. Raven, and M. Schofield, *The Presocratic Philosophers: A Critical History with a Selection of Texts,* 2d ed. (Cambridge, 1983), pp. 163–80 and 457; G. E. R. Lloyd, "Theories of Progress and Evolution," in *Civilization of the Ancient Mediterranean: Greece and Rome,* 3 vols., ed. Michael Grant and Rachel Kitzinger (New York, 1988), 1:265–75; and Michael J. O'Brien, "Xenophanes, Aeschylus, and the Doctrine of Primeval Brutishness," *Classical Quarterly* 35 (1985): 264–77.

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technical arts and crafts.[7] At least some contemporary Athenian philosophers incorporated that appreciation into their thought. Anaxagoras (ca. 500–ca. 428 B.C.), a philosopher who spent most of his active life in Athens, was a close associate of Pericles. He distinguished humans from other living beings because of their capacity (among others) for developing the arts and crafts.[8] The atomist Democritus (ca. 460–ca. 370 B.C.) elaborated a mechanistic view in which "necessity separated" certain arts indispensable for human existence. Arts such as weaving, building houses, and singing were discovered as humans imitated and became the pupils of animals. Of the more than sixty books that Democritus wrote, several concerned practical matters such as medicine, agriculture, painting, and warfare.[9]

Fifth-century B.C. anthropological accounts of the human acquisition of the arts would take backstage to the idealism of Plato and the teleological orientation of Aristotle. Yet the survival or reemergence of a view that humans acquired civilization through the gradual human discovery of the arts is attested by the later accounts of Lucretius, Diodorus, and Vitruvius in the 1st century B.C. and by a letter of Seneca (presumed to reflect the views of Posidonius), among others. Thomas Cole, enlarging a thesis first proposed by Karl Reinhardt in 1912, has carefully attempted to trace these later accounts back to Democritus.[10] For the purposes of this article, the sometimes controversial details concerning lines of influence are less important than the recognition of positive ancient traditions concerning the human acquisition of the arts. The centrality of the technical arts themselves for human civilization and the role of human inge-

[7]See esp. J. J. Pollitt, *Art and Experience in Classical Greece* (Cambridge, 1972), pp. 64–110.

[8]*Encyclopedia of Philosophy*, s.v. "Anaxagoras of Clazomenae," by G. B. Kerferd. For fragments of Anaxagoras's writings and testaments, see Diels and Kranz, eds. (n. 6 above), 2:5–44. Fragment 21b concerns the human capacity for the crafts. See also Edelstein, *Idea of Progress* (n. 6 above), pp. 22–23; and Kirk et al. (n. 6 above), pp. 352–57 and 459–60.

[9]*Encyclopedia of Philosophy*, s.v. "Leucippus and Democritus," by G. E. R. Lloyd. For the fragments, see Diels and Kranz, eds. (n. 6 above), 2:81–230, esp. frags. 144 and 154, which concern the arts. Essential studies include Karl Reinhardt, "Hekataios von Abdera und Demokrit," *Hermes* 47 (1912): 492–513; Gregory Vlastos, "Ethics and Physics in Democritus," *Philosophical Review* 54 (September 1945): 578–92, and 55 (January 1946): 53–64; and Vlastos, "On the Pre-history in Diodorus," *American Journal of Philology* 67 (1946): 51–59. See also Edelstein, *Idea of Progress* (n. 6 above), pp. 23–26; and Kirk et al. (n. 6 above), pp. 402–33 and 460.

[10]Thomas Cole, *Democritus and the Sources of Greek Anthropology* (Cleveland, 1967); Reinhardt, "Hekataios"; and for a critique of Cole's work, the review by Friedrich Solmsen in *Phoenix* 23 (1969): 399–402.

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nuity in their discovery are two views that found significant expression in the mid-to-late 5th century B.C. and thereafter in the ancient world.

Authorship and Invention in Ancient Practical Writings: The Example of Philo of Byzantium

Manuals devoted to practical, technical, and military subjects constitute further evidence for ancient positive traditions of practice. Initially the development of such writings was influenced by the Sophists, itinerant teachers of rhetoric who worked in and about Athens in the mid-to-late 5th century B.C. The Sophists attempted to explain the universe in terms of its phenomenal aspects alone, without appeal to outside principles or to a real world beyond phenomena. They were rebelling against the Eleatic tradition, which refused to recognize the phenomenal world as real. Although Sophistic thought was centered on issues of ethics, law, and morality, its view of phenomena as the locus of truth had positive implications for handwork and practice. The Sophist Protagoras (ca. 490–421 B.C.) promulgated a view of the origins of human society as the progressive development of the arts and crafts.[11]

The rich ancient traditions of practical and military writings, which developed under the initial influence of the Sophists, are important and neglected sources for attitudes toward authorship and invention.[12] Here it is possible to elaborate with only one example, albeit an important one, that of the Hellenistic inventor/author Philo of Byzantium (ca. late 3d century B.C.).[13] In a treatise on artillery

[11]For an introduction to the Sophists, see particularly George B. Kerferd, *The Sophistic Movement* (Cambridge, 1981); George B. Kerferd, ed., *The Sophists and Their Legacy*, Hermes, no. 44 (Wiesbaden, 1981); William K. C. Guthrie, *The Sophists* (1969; reprint, Cambridge, 1971), esp. pp. 60–68 on Protagoras's view of human progress; Rodolfo Mondolfo, *Il pensiero antico* (Florence, 1967), pp. 129–50; and Mario Untersteiner, *The Sophists*, trans. Kathleen Freeman (New York, 1954). For the fragments of the Sophists, see Diels and Kranz, eds. (n. 6 above), 2:252–416. For Protagoras, see also Antonio Capizzi, *Protagora: Le testimonianze e i frammenti* (Florence, 1955).

[12]Sophistic influence on the development of practical and military writings is suggested by the Illinois Greek Club [W. A. Oldfather et al.], *Aeneas Tacticus, Asclepiodotus, Onasander*, Loeb Classical Library (London, 1923), p. 7. An important study of ancient practical writings is Manfred Fuhrmann, *Das systematische Lehrbuch: Ein Beitrag zur Geschichte der Wissenschaften in der Antike* (Göttingen, 1960).

[13]For an introduction to Philo and his writings, see esp. *Dictionary of Scientific Biography*, s.v. "Philo of Byzantium," by A. G. Drachmann; A. G. Drachmann, *Ktesibios, Philon and Heron: A Study of Ancient Pneumatics* (Copenhagen, 1948), and *The Mechanical Technology of Greek and Roman Antiquity: A Study of the Literary Sources* (Copenhagen, 1963); Gian Arturo Ferrari, "Meccanica 'allargata,' " in *La scienza ellenistica*, Atti delle tre giornate di studio tenutesi a Pavia dal 14 al 16 Aprile 1982, ed. Gabriele Giannantoni

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machines, the *Belopoeica,* Philo specified his reasons for writing. His predecessors had used diverse methods. They differed concerning the proportions of the various parts of the machine and also with regard to the most important guiding factor, the hole that receives the spring. He therefore would ignore the old authors and use later sources only as their suggestions were effective in practice.[14]

Philo asked why some artillery engines were more effective than others, even though similar in materials and construction. Citing Polyclitus the sculptor that "perfection was achieved gradually in the course of many calculations," he elaborated that small discrepancies resulted in large total errors, and that special caution was required especially when changing scale. Philo illustrated his point with a discussion of how the proper calibration of artillery machines first was discovered:

> In the old days, some engineers were on the way to discovering that the fundamental basis and unit of measure for the construction of engines was the diameter of the hole. This had to be obtained not by chance or at random, but by a standard method which could produce the correct proportions at all sizes. It was impossible to obtain it except by experimentally increasing and diminishing the perimeter of the hole. The old engineers, of course, did not reach a conclusion . . . since their experience was not based on a sound practical foundation; but they did decide what to look for. Later engineers drew conclusions from former mistakes, looked exclusively for a standard factor with subsequent experiments as a guide, and introduced the basic principle

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and Mario Vegetti (Pavia, 1984), pp. 225–96; P. M. Fraser, *Ptolemaic Alexandria,* 3 vols. (Oxford, 1972), 1:428–34 and 2:619–20, nn. 425 and 426; Bertrand Gille, *Les mécaniciens Grecs: La naissance de la technologie* (Paris, 1980), pp. 103–21; F. Haase, "Philon," in *Allgemeine Encyklopaedie der Wissenschaften und Künste,* ed. Johann S. Ersch and J. G. Gruber (Leipzig, 1847), vol. 23, sec. 3 (O–Z).

[14]For the Greek text and English translation, see Eric W. Marsden, *Greek and Roman Artillery: Technical Treatises* (Oxford, 1971), pp. 105–84, p. 156, n. 1 (for previous editions), and pp. 106–9 for the portion of the text referred to here. Calibration and the proportional size of parts are discussed by Barton C. Hacker, "Greek Catapults and Catapult Technology: Science, Technology, and War in the Ancient World," *Technology and Culture* 9 (January 1968): 34–50, esp. 48–49. For the military context, see also Yvon Garlan, "Cités, armées et stratégie à l'époque Hellénistique d'aprés l'oeuvre de Philon de Byzance," *Historia* 22 (1973): 16–33, and *Recherches de poliorcétique Grecque,* Bibliothèque des Écoles Françaises d'Athènes et de Rome, 223 (Paris, 1974); and, finally, Eric W. Marsden, *Greek and Roman Artillery: Historical Development* (Oxford, 1969).

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of construction, namely the diameter of the circle that holds the spring.[15]

Philo emphasized that everything cannot be accomplished by the theoretical methods of pure mechanics. Much is to be found by experiment. He provided a second example that concerned the gradual experimental progress of architecture in which the proper proportions of buildings had been discovered by trial and error.[16]

Philo's notion of the progress of invention centers on ongoing experimentation. It assumes some kind of transmission and cooperation over time that results in gradual improvements and the discovery of a key factor (the diameter of the hole that holds the spring). The gradualism and the cooperative aspects of this view of the inventive process make it similar to the more general anthropological accounts of the human acquisition of the arts. Philo may well be indebted to those traditions.

Yet some attitudes familiar to the modern reader concerning authorship and invention are notably absent from Philo's account. There is no suggestion that superior ingenuity or the genius of any one particular individual is a factor in inventive progress. Nor does Philo suggest that the "old engineers" who made improvements and discoveries felt proprietary attitudes over them. There is no evidence that Philo himself had a particular interest in crediting the "old authors" although he may well have assumed that they were known to his own readers. Finally, although Philo is critical of the previous tradition on technical grounds, he does not appear to bestow on himself any special ingenuity beyond a place in the ongoing, progressive, and, by implication, cooperative tradition of technical advancement.

Authorship, Literary Theft, and Credit: The Roman Modification of the Hellenistic Tradition

The open, progressive, Hellenistic tradition of practical writing represented by Philo of Byzantium was modified by the Romans in the 1st century B.C. This Roman modification involved the placement of greater emphasis on the credit and honor due to past authors (primarily authors of writings). It also involved the explicit condemnation of the theft of writings. Ancient practices of authorship and book distribution provide the relevant context for understanding these Roman views.

[15]Marsden, *Technical Treatises*, pp. 106–9.
[16]Ibid.

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Jens Erik Skydsgaard has described lucidly one ancient method of writing a book. It involved reading (or more usually being read to) and excerpting relevant passages from previous books. Authors did their own writing by a process of selection and elaboration. Their procedures were facilitated by scribes who would copy passages on to clay tablets and then transfer them to papyrus rolls. The goal was to collect the best and most up-to-date information on a subject and to elaborate it when necessary.[17]

As E. J. Kenney has elaborated, for all of antiquity, once the first copy of a book had been made and distributed, its fate was beyond the author's control. There was no legal or practical way to safeguard the integrity of a text or limit the number of copies. Moreover, it was impossible to insure that an author's corrected or amended edition of a book would supersede the first distributed version. After its first distribution, it was not unusual for parts of a book to be excerpted into anthologies. It might also suffer adulteration in various other ways, including distribution under the name of a new "author." Neither Greek nor Roman laws included any notion of intellectual property. Yet accusations of theft or plagiarism were commonplace in both the Hellenistic and Roman worlds.[18]

Considering this steady reciprocal flow of theft and accusation, the views of the Roman architect/military engineer Vitruvius (fl. second half of 1st century B.C.) are notable. Vitruvius's treatise on architecture included treatment of materials, the design of temples, civic buildings, and private buildings, water supply, dials and clocks, and machines. His attitude toward past authors was one of veneration: "The ancients undertook not only wisely, but indeed also usefully to transmit their thoughts to posterity through reports of commentaries, so that they should not be lost; but increasing from generation to generation, having been published in books, step by step they arrived in a very long time at the highest subtlety of learning. Thus, not modest but infinite thanks must be given to these because they did not

[17]Jens Erik Skydsgaard, *Varro the Scholar: Studies in the First Book of Varro's "De re rustica,"* Analecta Romana Instituti Danici, vol. 4, Suppl. (Copenhagen, 1968), pp. 101–16.

[18]E. J. Kenney, "Books and Readers in the Roman World," in *The Cambridge History of Classical Literature,* vol. 2, *Latin Literature,* ed. E. J. Kenney and W. V. Clausen (Cambridge, 1982), pp. 3–32. For accusations of theft or plagiarism, see Eduard Stemplinger, *Das Plagiat in der griechischen Literatur* (Leipzig, 1912); and Konrat Ziegler, "Plagiat," *Paulys Real-encyclopädie der classischen Altertumswissenschaft* 20.2 (1950): cols. 1956–97. See also Victor Hazan, "The Origins of Copyright Law in Ancient Jewish Law," *Bulletin of the Copyright Society of the U.S.A.* 18 (1970–71): 23–28, who has described the belief in correct attribution of written and spoken words rather than the origins of copyright law per se.

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enviously neglect the dead, but attended to transmitting for memory notions of all kinds through writings."[19] In this remarkable passage, the ancients are honored not only for writing down their own ideas, but also for transmitting the thoughts of their predecessors.

Vitruvius took care to credit authors, inventors, and particularly inventors who were also authors. He provided a long list of his own sources, naming authors of writings on stage scenery, the proportions of the various architectural orders, the laws of symmetry, and machines, among other subjects.[20] He also insisted that he himself did not write by inserting his own name "after changing the titles of other men's books." Nor was his intention "to win approbation by finding fault with the ideas of another." Rather, he expressed "unlimited thanks to all the authors that have in the past, by compiling from antiquity remarkable instances of the skill shown by genius, provided us with abundant materials of different kinds. Drawing from them as it were water from springs, and converting them to our own purposes, we find our powers of writing rendered more fluent and easy, and, relying upon such authorities, we venture to produce new systems of instruction."[21]

The other side of Vitruvius's esteem for authorship was his harsh condemnation of the theft of writings. Those who, "stealing the writings" of other authors, "publish them as their own" must be censored. Further, those who "rely in writings not on their own notions, but boast with odious behavior doing violence to the works of

[19]Vitruvius, *De architectura,* ed. and trans. Frank Granger, Loeb Classical Library (Cambridge, Mass., and London, 1931), 7.preface, 1. I am indebted to the translations of Granger and of Morris Hickey Morgan, *Vitruvius: The Ten Books on Architecture* (1914; reprint, New York, 1960) but have often retranslated the Latin. Scholarship on the text is now facilitated by L. Callebat, P. Bouet, P. Fleury, and M. Zuinghedau, eds., *Vitruve, De architectura: Concordance, documentation bibliographique, lexicale et grammaticale,* 2 vols. (Hildesheim, 1984). What we know about Vitruvius, who served as a military engineer under Julius Caesar, is limited to references to his life in his treatise—*De architectura,* 1.preface, 2, 5.1.6–10, and 6. preface, 3–4. The thesis of Paul Thielscher, "Vitruvius," *Paulys Real-encyclopädie der classischen Altertumswissenschaft* 17.2 (1961): cols. 419–89, that Vitruvius was actually Mamurra, Julius Caesar's *praefectus fabrum* in Gaul, was decisively refuted by P. Ruffel and J. Soubiran, "Vitruve ou Mamurra?" *Pallas* 11 (1962): 123–79. Two recent comprehensive bibliographic essays are Pierre Gros, "Vitruve: L'architecture et sa théorie à la lumière des études récentes," and Louis Callebat, "La prose du 'De Architectura' de Vitruve," in *Aufstieg und Niedergang der Römischen Welt. II. Principat,* vol. 30.1 (1982), pp. 659–95 and 696–722, respectively. On Vitruvius's place in scientific and technical culture, see Giorgio Tabarroni, "Vitruvio nella storia della scienza e della tecnica," *Atti della accademia delle scienze dell'istituto di Bologna, Classe di scienze morali, Memorie* 66 (1971–72): 1–37.

[20]Vitruvius, *De architectura,* 7.preface, 11–18.

[21]Ibid., 7.preface, 10. Morgan's translation is cited here (n.19 above), pp. 197–98.

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others, who have lived in an impious way, not only must be censored, but must be condemned to punishment."[22] The Roman architect elaborated with a story about Aristophanes of Byzantium (ca. 257–180 B.C.), the literary scholar and librarian of the famous library in Ptolemaic Alexandria. In a poetry contest sponsored by the king, Aristophanes discovered that all the popular poets had stolen "their" poems from the collections in the library. He declared the least popular (but only original) poet the winner and calmed the indignant king by proving the thefts of the others after fetching the appropriate volumes. The original poet was thereby rewarded (as was Aristophanes), whereas the poetry thieves were condemned and dismissed.[23]

But what was the precise significance of the stolen object in an ancient literary theft? There is little evidence that either writings or inventions were viewed as commodities with a marketplace value in the ancient world. Vitruvius himself insisted that he had not studied to make money from skill. He believed that small means with good repute *(bona fama)* was better than wealth and infamy. Thus little celebrity had followed. Nevertheless, he hoped that "by publishing these volumes, I will be known even to posterity." Elsewhere, Vitruvius lamented the honors bestowed on athletes, while authors were ignored. After describing the contributions of particular ancient authors, he specified their true compensation: "Although honors are not bestowed on writers either for [their] excellent characters or teachings, nevertheless their minds gazing by themselves at the higher region, are raised to heaven by the steps of remembrance; they compel that not only their opinions but even their appearances are known to posterity for immortal time."[24] Authorship, in short, brings immortal fame, and it is that precious legacy that is lost in literary theft.

Respect for past authorship continued to be characteristic of Roman writings in the 1st century A.D. An important example is Pliny the Elder (A.D. 23/24–79), whose only surviving work, *Naturalis historia,* contains 37 books treating the universe, geography, man, other animals, botany, agriculture, metals and stones, art, architec-

[22]Ibid., 7.preface, 3.

[23]Ibid., 7.preface, 4–7. See P. M. Fraser, "Aristophanes of Byzantion and Zoilus Homeromastix in Vitruvius: A note on Vitruvius VII, Praef. 4–9," *Eranos* 68 (1970): 115–22, for a detailed analysis of this passage.

[24]Vitruvius, *De architectura,* 6.preface, 5, and 9.preface, 1–18, and 9.preface, 16 for citation. Jean Soubiran, *Vitruve: De l'architecture livre IX* (Paris, 1969), pp. xix–xxii, points out that criticism of the disproportionate rewards of athletes compared to thinkers and writers was a common theme in antiquity from the time of Xenophon.

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ture, and numerous other topics.[25] Pliny insisted that authors should receive credit for their work. He reported his discovery of numerous thefts by his own contemporaries:

> You will deem it a proof of this pride of mine that I have prefaced these volumes with the names of my authorities. I have done so because it is, in my opinion, a pleasant thing and one that shows an honourable modesty, to own up to those who were the means of one's achievements, not to do as most of the authors to whom I have referred did. For you must know that when collating authorities I have found that the most professedly reliable and modern writers have copied the old authors word for word, without acknowledgement. . . . Surely it marks a mean-spirit and an unfortunate disposition to prefer being detected in a theft to repaying a loan—especially as interest creates capital [*sors fiat ex usura*].[26]

Pliny's views on authorship and plagiarism presupposed a cumulative, progressive view of knowledge. Interest creates capital—that is, if one repays the interest on the loan of openly transmitted writings by acknowledging the debt, an even greater supply of knowledge will result.

Although Pliny's acknowledgment of past authorship included the desire to assign due credit, it was also influenced by considerations of encyclopedism. In his book on humans, for example, he provided the names of the inventors and discoverers of the arts and sciences and of tools and methods. The sequence of his list is instructive. His

[25]I have used Pliny, *Natural History*, ed. H. Rackham, 10 vols., Loeb Classical Library (Cambridge, Mass., 1938–63). An introduction to the large literature on Pliny the Elder and his encyclopedia can be found in Rudolf Hanslik, "Der Forschungsbericht: Plinius der Ältere," *Anzeiger für die Altertumswissenschaft* 8 (October 1955): cols. 193–218, and "Der Forschungsbericht: Plinius der Ältere, II. Bericht," *Anzeiger für die Altertumswissenschaft* 17 (April 1964): cols. 65–80; Franz Römer, "Der Forschungsbericht: Plinius der Ältere, III. Bericht," *Anzeiger für die Altertumswissenschaft* 31 (July–October 1978): cols. 129–206; Klaus Sallman, "Plinius der Ältere: 1938–1970," *Lustrum* 18 (1975): 5–299 and 344–55; and Guy Serbat, "Pline l'Ancien: État présent des études sur sa vie, son oeuvre et son influence," *Aufstieg und Niedergang der Römischen Welt. II. Principat*, vol. 32.4 (1986), pp. 2069–2200. For a particularly valuable group of essays on Pliny's science and technology, see Roger French and Frank Greenaway, eds., *Science in the Early Roman Empire: Pliny the Elder, His Sources and Influence* (Totowa, N.J., 1986).

[26]Pliny, preface, 22–23. For a detailed analysis of the entire preface (which consists of a dedicatory letter to the future Emperor Titus, son of Vespasian), see T. Köves-Zulauf, "Die Vorrede der plinianischen 'Naturgeschichte,' " *Wiener Studien*, neue Folge 7 (1973): 134–84.

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inventory of eminent men in science, medicine, and art[27] is followed by a discussion of exceptional prices that have been paid for slaves.[28] Examples of the happiest humans[29] and of unusual happenings demonstrating the vicissitudes of fortune[30] are followed by a list of the discoverers of the arts and sciences (many of them mythical) and inventors of weapons.[31] In his discussion of founders and inventors, Pliny is not only giving credit to the great inventors and thinkers of the past but is also filling an encyclopedia that focuses on the exceptional rather than the average.

Thus far in the discussion of both Hellenistic and Roman practical authorship, some of the elements of a concept of intellectual property seem to be in evidence. Crediting of authorship and the Roman insistence on honoring past authors may well be components of the type of intellectual property that is grounded in original authorship. Complaints concerning theft are also tied to the honor of the author. Yet the complete lack of a concept of intellectual property in ancient law should caution against the assumption that concern with credit, honor, and theft add up to a concept of intellectual property. I would suggest that they do not. Two essential elements are missing. The first is the explicit separation of tangible from intangible aspects of a work. The second is the notion that the intangible products of authorship consist of commercial property. The ancient theft of writings constituted a purloining of honor, fame, and reputation. It was never considered larceny of intangible commercial property.

Craft Secrecy: Problems of Definition and Evidence

Despite the importance of practical and technical writings, it is still true that in the ancient world and well beyond, most technical and

[27]Pliny, 7.37–38. For a cogent discussion of the entire Book 7 on humans, see Zoltán Kádár and Maria Berényi-Révész, "Die Anthropologie des Plinius Maior," *Aufstieg und Niedergang der Römischen Welt. II. Principat*, vol. 32.4 (1986), pp. 2201–2224. For Pliny's encyclopedism, see Pierre Grimal, "Encyclopédies antiques," *Cahiers d'histoire mondiale* 9 (1966): 477–82; and Angelo Roncoroni, "Plinio enciclopedista," in *Plinio e la natura*, Atti del ciclo di conferenze sugli aspetti naturalistici dell'opera pliniana (Como, 1982). The tradition of identifying inventors was in evidence in the 5th century B.C. and continued to the early modern period. See especially Klaus Thraede, "Das Lob des Erfinders: Bemerkungen zur Analyse der Heuremata-Kataloge," *Rheinisches Museum für Philologie* 105 (1962): 158–86; and Brian P. Copenhaver, "The Historiography of Discovery in the Renaissance: The Sources and Composition of Polydore Vergil's 'De inventoribus rerum,' I–III," *Journal of the Warburg and Courtauld Institutes* 41 (1978): 192–214.

[28]Pliny, 7.39.

[29]Ibid., 7.40.

[30]Ibid., 7.41–7.45.

[31]Ibid., 7.56–7.60.

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craft knowledge was transmitted orally through apprenticeship systems. However, concrete evidence for craft secrecy is sparse. Where craft secrecy did exist, can we assume that it constituted a form of protection for "intellectual property"? Scholarly references to "craft secrets" are as frequent as they are unexamined.

An automatic correlation between the oral transmission of craft knowledge and secrecy should not be assumed. Many crafts at one time or another constituted the required skills of household production. As urbanism developed, artisan production tended to become separated from the household. Yet ongoing household production of craft items would necessarily have been far higher in ancient times, and indeed in all premodern eras, than it is today. For most ancient households and rural estates, the acquisition of the knowledge and skill of a particular craft would have depended on gender and class, but would not necessarily have been the result of a particularly secret course of instruction. Most ancient people would have been far more familiar with craft production than their modern counterparts, even when their social class might encourage them to disdain the value of handwork.[32]

On the basis of some evidence, we can assume that craft skills often were handed down through families, from father to son, and from mother to daughter, whether within the context of households, craft guilds, or some other organization. With reference to Greece and Rome, Alison Burford has confirmed that "craft secrets" often were handed down within families from one generation to the next. Burford points both to literary references and to personal records which show that fathers often trained their sons in the crafts into which they had been born. The family line of transmission is emphasized by the custom of craftsmen adopting young relatives or friends as both apprentices and heirs when they lacked offspring of their own to fill such roles.[33] In some cases at least, craft apprenticeship may have involved learning specifically secret information. Thus

[32]For instance, Alison Burford, *The Greek Temple Builders at Epidauros: A Social and Economic Study of Building in the Asklepan Sanctuary, during the Fourth and Early Third Centuries B.C.* (Toronto, 1969), esp. p. 198, suggests that the Epidaurans were "unused to specialists" and more familiar with "occupations basic to maintaining a decent standard of living, stone masonry and woodwork."

[33]Alison Burford, *Craftsmen in Greek and Roman Society* (Ithaca, N.Y., 1972), pp. 82–91, where a range of evidence, mostly from Greek sources, is presented. On the other hand, in Burford's detailed study of a particular building site, *Greek Temple Builders*, esp. pp. 222–36, the evidence for family relationships among skilled craftsmen is slight. Family connections are much more in evidence among the administrators. For a good summary of ancient craftsmen, see Alison Burford, "Crafts and Craftsmen," in Grant and Kitzinger, eds. (n. 6 above), pp. 367–88.

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Xenophon contrasted farmers who were open with their knowledge with artisans who "conceal more or less the most important points in their own art."[34]

Yet concrete evidence for craft secrecy in antiquity is exceedingly sparse.[35] The possible meanings of such secrecy, where it did exist, need further scrutiny. The philosopher Sissela Bok has provided a useful definition of secrecy as "intentional concealment" and has distinguished it from unknown things (e.g., the secrets of nature) and privacy ("the condition of being protected from unwanted access by others").[36]

The one form of "craft secrecy" common to all crafts comes under the category of "unknown things" in Bok's analysis. That is, for every craft, there is a significant body of knowledge that can only be "known" through actually practicing the craft with one's own hands. Written or spoken instructions can introduce the subject but cannot actually transmit it. Only through learning the technique itself through practice can one truly know it. The "secrets" of these crafts are synonymous with the "techniques."[37]

When craft secrecy is "true" secrecy involving intentional concealment, the motivation for that concealment is intrinsic to its meaning.

[34]Xenophon, *Memorabilia and Oeconomicus,* trans. E. C. Marchant, Loeb Classical Library (London, 1938), *Oeconomicus,* 15.11–12.

[35]To the objection that there is just no evidence for what may have been a common practice, I can only suggest that if craft secrecy based on a view of proprietary knowledge were widespread in antiquity, there would be some evidence for it. The discussion of Finley (n. 5 above), pp. 29–45, concerning the lack of modern economic notions in the ancient world, is relevant here. It is likely that scholars have projected onto ancient craft contexts what they know about medieval craft practice (where evidence for craft secrecy is extensive).

[36]Sissela Bok, *Secrets: On the Ethics of Concealment and Revelation* (New York, 1983), pp. 5–14.

[37]The use of "secret" to mean "technique" remains in contemporary English usage. Thus the "secret" of making good pie crust is to handle the dough lightly and use cold water. (I refer to a technique, not to information that has been concealed intentionally.) Likewise, late medieval "books of secrets," whatever else they represented, were books of techniques. Their early publication in the 16th century may have involved the disclosure of information that was formerly "intentionally concealed," but may also be seen as the printed dissemination of well-known techniques. On the books of secrets tradition, see esp. William Eamon, "Arcana Disclosed: The Advent of Printing, the Books of Secrets Tradition and the Development of Experimental Science in the Sixteenth Century," *History of Science* 22 (1984): 111–50. The distinction between the secret as technique and the secret as that which is intentionally concealed is also germane to the valuable discussions of Lon R. Shelby, "The 'Secret' of the Medieval Masons," in *On Pre-Modern Technology and Science: Studies in Honor of Lynn White, jr.* (Malibu, Calif., 1976), and *Gothic Design Techniques: The Fifteenth-Century Design Booklets of Mathes Roriczer and Hanns Schmuttermayer* (Carbondale, Ill., 1977), pp. 46–61.

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Anthropological studies of apprenticeship have shown that craft secrets can refer to social secrets and to the "lore" of a craft as much as to technical secrets. Such secrets help to ensure the privileged status of the master-apprentice relationship and of the craft itself.[38] Studies by social scientists have clarified the role of secrecy within interpersonal relationships and groups.[39] If craft secrets involve technical secrets, they could function by enhancing the status of the practitioner. If a novice asks a potter how she makes her beautiful pots, she could tell him in explicit detail and even provide demonstrations. On the other hand, she could be intentionally vague, withholding key information. The novice might then believe that pot making was a mystery requiring special talents that she had, but he did not. Making pots becomes not just a matter of practice, but a craft requiring mysterious endowments that enhance the prestige of those who have them. Likewise, craft secrecy could enhance the insider status and cohesiveness of a craft group vis-à-vis outsiders.

A very different motivation for craft secrecy posits craft and technical knowledge as intangible property that has commercial value apart from the products the artisan can produce as a result of such knowledge. Craft secrecy in such a case involves the protection of intellectual property. My own studies of ancient guild documents, craft recipes, practical manuals, and other sources have turned up no evidence for such a concept in the world of ancient practice.

Did Ancient Practitioners Hold Proprietary Attitudes toward Craft Knowledge? The Negative Evidence of Ancient Guilds

An examination of the evidence for ancient guilds underscores the fact that the artisans of antiquity operated within value systems profoundly different from those involving more modern economic or intellectual property assumptions. The development of guilds—organizations of artisans based on particular crafts and technologies—occurred independently in many areas of the world and in many different periods of premodern history. Although the appearance of guilds seems to presuppose a certain development of the crafts

[38]See esp. R. M. Dilley, "Secrets and Skills: Apprenticeship among Tukolor Weavers," in *Apprenticeship: From Theory to Method and Back Again*, ed. Michael W. Coy (Albany, N.Y., 1989), pp. 181–98, who describes the most important secrets of a group of West African weavers not as weaving techniques but as weaving "lore," including the sixteen names of the mythical ancestor of the weavers, the line of fictive descent from the weaver to the ancestor, myths and legends associated with the craft, weaving origin myths and their relationship to contemporary practice, and verses, incantations, and spells.

[39]See esp. Stanton K. Tefft, ed., *Secrecy: A Cross-cultural Perspective* (New York, 1980), which includes extensive further bibliography.

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beyond a household industry as well as some degree of urbanism, these two conditions did not invariably lead to their genesis. When guilds did emerge, it is clear that they were not primarily economic organizations, nor were they concerned with the maintenance of secrecy within a particular craft.[40]

There is little evidence for widespread guild activity in classical Greece or republican Rome. Far greater notice of craft guilds occurs in the Hellenistic period. Craft association became commonplace throughout the Roman territories only from the time of the Emperor Augustus (28 B.C.–A.D. 14). In the 4th century A.D. the guilds of the empire were transformed from voluntary associations with government approval into "compulsory public service corporations entirely controlled by the state," the members coming "to constitute one of the hereditary castes into which the population of the late empire was divided." With the decline of urbanism in the West, guilds for the most part disappeared. Many scholars believe that medieval merchant and craft guilds arose independently with the redevelopment of the cities in the 11th century.[41]

Ancient craft guilds were not generally involved in the training of apprentices, the setting of standards, or the protection of craft secrets. They seem to have begun as free associations of artisans, or of owners and managers of artisan industries. Sometimes slaves and freemen joined together, sometimes they included women. Often the primary purpose of the guild was to attend to the cult of a deity, or to provide burial for poor members, or to meet for social purposes at feasts and banquets. Being associations of the lowborn—artisans, laborers, and

[40]For an introduction to premodern guilds with the older bibliography, see *Encyclopaedia of the Social Sciences,* s.v. "Guilds," by Mariano San Nicolò et al.

[41]Ibid., esp. pp. 206–7 ("Late Roman and Byzantine," by A. E. R. Boak) for citations. For further bibliography, see esp. Francesco M. De Robertis, *Il fenomeno associativo nel mondo romano dai collegi della repubblica alle corporazioni del basso impero* (Naples, 1955); Patrick W. Duff, *Personality in Roman Private Law* (Cambridge, 1938), pp. 95–158; Theodor Mommsen, *De collegiis et sodaliciis Romanorum* (Kiel, 1843); *Oxford Classical Dictionary,* 2d ed., s.v. "Clubs, Greek," by Marcus Niebuhr Tod, and s.v. "Clubs, Roman," by George Hope Stevenson; Franz Poland, *Geschichte des griechischen Vereinswesens,* Preisschriften gekrönt und herausgegeben von der Fürstlich Jablonowskischen Gesellschaft zu Leipzig, 38 (Leipzig, 1909), esp. pp. 116–27; *Paulys Real-encyclopädie der classischen Altertumswissenschaft,* new ed., 4 (1901): cols. 380–480, s.v. "Collegium," by Ernst Kornemann; *Paulys Real-encyclopädie der classischen Altertumswissenschaft,* new ed., suppl. 4 (1924): cols. 155–211, s.v. "Berufsvereine," by Albert Stöckle; Max Radin, *The Legislation of the Greeks and Romans on Corporations* (New York, 1910); Mariano San Nicolò, *Ägyptisches Vereinswesen zur Zeit der Ptolemäer und Römer,* 2 vols. (Munich, 1913–15), esp. 1:67–206; Marcus N. Tod, *Sidelights on Greek History* (Oxford, 1932), pp. 71–96; and Jean-Pierre Waltzing, *Étude historique sur les corporations professionnelles chez les Romains,* 4 vols. (1895–1900; reprint, New York, 1979).

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slaves—they are rarely mentioned in literary texts. On the other hand, a massive number of inscriptions refer to one or another of them. In addition, papyrus records and legal codes provide some information. Nevertheless, the evidence for each association that we know about is fragmentary and limited and usually reveals more about internal organization than about activities and goals.[42]

The guilds of the Romans (*conlegia* in the republic, *collegia* in the imperial period) can illustrate some of the usual features of ancient guilds. Historians generally agree that Plutarch's description of the founding of the Roman *collegia* by King Numa is false, being too early. The actual origin of these associations of artisans is obscured by a virtually total lack of evidence. The assessment of Jean-Pierre Waltzing, whose study remains fundamental, is that the Roman guilds were originally private, free, religious associations that included the cult of a deity and provided burial for impecunious members. They also could be manipulated for political ends as they were in the tumultuous days of the Cataline conspiracy (64 B.C.). It was then that the Roman government attempted their suppression or regulation, an effort that was first successful under Augustus. If the association was not viewed as a threat to the social and political order, authorization was given in exchange for some public service.[43]

In contrast to medieval artisan guilds, ancient guilds usually did not have the economic functions of protecting a craft, perfecting and conserving craft techniques, or providing for and regulating apprenticeship. With a few exceptions, guilds did not exercise monopolies over trades and industries. The membership of a *collegium* did not usually contain all the workers of a particular craft, and sometimes it consisted of the workshop owners and managers rather than the actual laborers. Membership was not compulsory. Yet the associations did sometimes act to defend or advance the material interests of

[42]Francesco M. De Robertis, *Storia delle corporazioni e del regime associativo nel mondo romano,* 2 vols. (Bari, 1973); and Waltzing, 1:161–233. See also Russell Meiggs, *Roman Ostia,* 2d ed. (Oxford, 1973), pp. 311–36, for an excellent discussion of the guilds of a particular locality.

[43]Waltzing, 1:61–78, and for their suppression in 64 B.C., 1:90–113. See also Francesco M. De Robertis, *Il diritto associativo Romano dai collegi della repubblica alle corporazioni del basso impero* (Bari, 1938), pp. 3–240; Lily Ross Taylor, *Party Politics in the Age of Caesar* (Berkeley and Los Angeles, 1949), p. 44; and Susan Treggiari, *Roman Freedmen during the Late Republic* (Oxford, 1969), pp. 168–77. The extent and nature of government control in the late republic is debated by specialists. See, e.g., Radin (n. 41 above), pp. 91–97, who interprets the *lex Julia Augusti* as referring to the suppression of brigands, rather than *collegia* in general. For Plutarch's discussion, see Bernadotte Perrin, trans., *Plutarch's Lives,* 11 vols., Loeb Classical Library (Cambridge, Mass., 1982), 1:365–67 (*Numa,* 17).

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members vis-à-vis the state, whether or not the matter concerned the craft itself. Many *collegia* looked to one or more private patrons to defend the interests of their members. Nevertheless, the primary functions of Roman artisan associations, as indeed of all ancient craft guilds, were religious, social, funerary, or more often than not a combination of the three, rather than economic.[44]

In the early empire many craft associations took on public services that came to be seen as obligations. From the time of the early principiate, the government depended on private associations to carry out the provisioning of the city of Rome. During the 3d century, state control over the associations was established. The emperor Severus Alexander (A.D. 222–235) in an expanding autocracy appointed judicial representatives from each guild, placing them under the jurisdiction of certain courts. Henceforth guilds operated under government supervision as part of the administration. During the 4th and 5th centuries, associations (now called *corpora*) were charged with increasingly onerous obligations, failure of which resulted in the confiscation of individual members' property. The associations became fully branches of the state; first the provision of goods and services and then hereditary membership became compulsory. Members tried to escape their burdens often by changing profession or location. Private social, religious, and funerary functions disappeared, in part because the spread of Christianity absorbed some of these functions, in part in consequence of the change from private free clubs with a public obligation to state organizations with all-encompassing burdens imposed by an autocratic monarchy.[45]

Had ancient artisans been significantly motivated by proprietary attitudes toward their own craft skill and their inventions, it is reasonable to believe that the evidence of ancient guilds, fragmentary though it is, would provide evidence for it. Ancient artisans undoubtedly took pride in their skills. But there are few grounds for supposing that they saw their craft knowledge, their skill, the processes they used, or the novelties they developed specifically as possessions to be protected or exploited aside from the products they made or the wages or other livelihood they earned. Put in another way, evidence for the concept of intellectual property as

[44]Waltzing, 1:161–333.

[45]Arthur E. R. Boak, *Manpower Shortage and the Fall of the Roman Empire in the West* (Ann Arbor, Mich., 1955), pp. 64–84; Arthur E. R. Boak and William G. Sinnigen, *A History of Rome to A.D. 565* (New York, 1965), pp. 348–51, 375, and 464–65; De Robertis, *Fenomeno associativo* (n. 41 above), pp. 137–204; Samuel Dill, *Roman Society in the Last Century of the Western Empire,* 2d rev. ed. (New York, 1958), pp. 227–81; Waltzing, 2:245–348.

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distinct from the property of goods and wages is lacking for ancient craft contexts.

Authorship and Practice in Late Antiquity and the Middle Ages

To investigate attitudes toward inventiveness and authorship in late antiquity and the early medieval era, it is necessary to turn again to diverse traditions of practical writing. One group of such writings, the alchemical, was to become the basis for the most important esoteric tradition of practice in the West. Authors of other kinds of practical writings viewed authorship within an exoteric framework.

Within alchemy the relationship of the knower to knowledge itself constituted an inversion of the relationship associated with "intellectual property." Instead of possessing and owning knowledge, alchemical initiates themselves became possessed and transformed by it. The disciple followed a master who taught the arts of both material and spiritual transformation. A passage in an early alchemical text, *Physica et Mystica* by pseudo-Democritus, tells of the students following precious instructions before and after the master's death. Providing instructions for making purple dye, the author notes that they were received from "our master" and that having recognized the diversity of matter "we are obliged to harmonize their natures." However, the master had died before they "were initiated," while they were "still occupied with the knowledge of matter." Having invoked him from Hades, the adept were instructed from beyond the grave that "the books are in the temple." Searching within, they found no books. But a column opened and the master's son led the pupils there where they found a "precious formula written: 'Nature rejoices in nature, nature triumphs over nature, nature dominates nature.' "[46] The remainder of the text concerns the illustration of this tripartite formula through a series of craft recipes.

From this illustration it can be seen that, for some, alchemy must have offered a deeply satisfying path. It involved small groups of people involved in intense master-student relationships whose cohesiveness would have been magnified by political marginality and suppression. It offered the mysteries of the laboratory with its wonderful substances and apparatus, where the initiated could not only enjoy the pleasures of handwork but could change the nature of matter itself. And it offered the steps to spiritual transformation paved with esoteric and difficult philosophical concepts and proce-

[46]M. Berthelot and C.-É. Ruelle, *Collection des anciens alchimistes grec,* 3 vols. (Paris, 1887–88), 3:44–45. See also the discussion and partial English translation of John Maxson Stillman, *The Story of Alchemy and Early Chemistry* (1924; reprint, New York, 1960), pp. 154–57.

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dures, guided by a master toward divine wisdom. Alchemical books recorded ancient wisdom; the true author was the Egyptian god Toth. Alchemical writings were often credited to well-known ancient authorities, a practice that enhanced their function as sacred guides.[47] Seeking the transmutation of both matter and spirit, the alchemists perceived that the acquisition of knowledge involved the transformation of the self. Instead of possessing knowledge, they were possessed and transformed by it.

Late antique and medieval authors of practical manuals wrote within exoteric traditions that stand in marked contrast to that of the alchemists. The writings of three authors—Vegetius in late antiquity, Eraclius in the 10th century, and Theophilus in the 12th—can be used to exemplify a variety of attitudes toward practical knowledge and authorship. It will be seen that these authors are remote indeed from concern with technical innovation and that they do not hold proprietary attitudes toward practical and technical knowledge. Yet, different as they are from one another, they share a fundamental concern for making traditional knowledge available for direct use in contemporary practice.

The most widely copied and influential military treatise in the medieval period was Vegetius's *Epitoma rei militaris*. Vegetius wrote the work sometime between A.D. 383 and 450 and addressed it to an emperor. Specialists continue to disagree on the precise year of publication and the particular emperor for whom it was written. Yet the general context of the late Roman Empire is clear, as are Vegetius's reformist and practical aims.[48]

Walter Goffart has described Vegetius's authorship as motivated by neither historical nor antiquarian concerns but rather by an intense interest in contemporary military practice. Vegetius's main goal was to

[47]See Robert Halleux, *Les textes alchimiques* (Turnhout, 1979); Christoph Meinel, ed., *Die Alchemie in der europäischen Kultur- und Wissenschaftsgeschichte* (Wiesbaden, 1986), for recent scholarship; and Robert P. Multhauf, *The Origins of Chemistry* (New York, 1966), esp. pp. 82–116 for a good introduction to early alchemy and p. 90 for alchemical authorship.

[48]Vegetius was possibly a government official and large landowner. In addition to his military treatise, he wrote a book on veterinary medicine, the *Mulomedicina*. The critical edition of his military treatise is *Flavi Vegeti Renati Epitoma rei militaris*, ed. C. Lang (1885; reprint, Stuttgart, 1967). See also Flavius Vegetius Renatus, *Epitoma Rei Militaris*, ed. and trans. Leo F. Stelton (New York, 1990). For the dating, see Walter Goffart, "The Date and Purpose of Vegetius' 'De Re Militari,' " *Traditio* 33 (1977): 65–100, and his *Rome's Fall and After* (London, 1989), pp. 45–80 and 155 (a reprint of the article with additions); and Foster Hallberg Sherwood, "Studies in Medieval Uses of Vegetius' *Epitoma rei militaris*" (Ph.D. diss., University of California, Los Angeles, 1980), pp. 11–19.

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elaborate ancient (Roman) military precepts that would have practical utility in the solution of the problems of the contemporary army.[49] He believed that instruction in arms should be demonstrated daily through practice, but that this had been rendered impossible by the long period of peace that had destroyed personal knowledge of these skills. Since no one can teach what he himself did not learn, it was necessary to search for ancient custom in books. He intended to bring together Roman writings, including those of Cato the Elder, Cornelius, Celsus, Frontinus, and Paternus, as well as the ordinances of Augustus, Trajan, and Hadrian. He described his own treatise as a carefully composed and organized epitome of these older writings.[50] Vegetius insisted on the value of authorship, noting that what has been lost from practice can be recovered from books.[51]

Vegetius treated the recruiting and training of soldiers and the organization, tactics, strategy, and artillery machines used in the siege of walled cities. Yet he declined to discuss technologies and other matters that had been improved after his sources had been written. He omitted cavalry because it had been perfected after the time of ancient writers,[52] and he refrained from elaborating on the sling-staff, arcuballista, and sling since they were well known from contemporary usage.[53] Concerning the Danube watch ship (*lusoria*) he wrote: "I believe I must be silent, because very frequent practice in these [boats] reveals more ample skill than the old instruction has taught."[54] In no sense then did Vegetius attempt to improve technology through his writings as Philo had. Neither was his aim to honor the old authors, although he believed they had lived in times superior to his own. He thought, rather, that old books contained knowledge lost to contemporary practice and he mined them according to the criteria of present military needs. Contemporary practice, even when it included novel inventions, was well known and thus not worth writing about.

Like Vegetius, the 10th-century author Eraclius emphasized the value of knowing prior practice. Eraclius elaborated his views toward authorship in the preface of *De coloribus et artibus Romanorum,* a collection of verses offering instruction on pigmentations, polishing

[49]Goffart, "The Date," pp. 88–100. This view is supported by the evidence of Vegetius's text, regardless of its precise dating. See also Bernard S. Bachrach, "The Practical Use of Vegetius' *De re militari* during the Early Middle Ages," *Historian* 47 (1985): 239–55.

[50]Vegetius, 1.8.

[51]Ibid., 3.10.

[52]Ibid., 3.26.

[53]Ibid., 4.22.

[54]Ibid., 4.46.

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gems, cutting crystal, and copper gilt: "I indeed write nothing to you, which I have not first tried myself. The greatness of intellect, for which the Roman people was once so eminent, has faded, and the care of the wise senate has perished. Who can now investigate these arts? Who is now able to show us what these artificers, powerful by their immense intellect, discovered for themselves? He who, by his powerful virtue, holds the keys of the mind, divides the pious hearts of men among the various arts."[55] Having emphasized that he himself had tested all the craft instructions, Eraclius praised the ingenuity of the ancient Romans. In his own time, God had divided knowledge of the arts among men.

The same spirit of humility is evident in the 12th-century treatise on painting, glass, and metalwork *De diversis artibus*. The pseudonymous author, Theophilus, states that he is "a humble priest, servant of the servants of God, unworthy of the name and profession of monk." Having described the creation and the fall, Theophilus emphasized that, although humans lost immortality, they still transmitted to posterity their knowledge and intelligence so that anyone who concentrated could acquire "the capacity for the whole range of art and skill." Once acquired, such skills should not be hidden "in the purse of envy" nor concealed "in the storeroom of a selfish heart." Rather, without boasting, the artisan would "simply and with a cheerful mind dispense to those who seek." Theophilus offered his own writing as an example of the way to proceed: "I, an unworthy human creature, almost without name, offer freely to all who desire in humbleness to learn the gifts that God . . . has deigned to grant freely to me." Urging his readers to read with "tenacious memory" and "embrace" his treatise "with ardent love," he explained that he had not written out of desire for praise or temporal reward, that he had "neither stolen anything precious or rare nor silently reserved anything" for himself alone. He wrote to give "aid to many men in their need" and was concerned for human advancement, for such would increase the glory of God's name.[56]

[55]Although Eraclius probably wrote on the Italian peninsula, nothing else about him is known. See Mary P. Merrifield, *Original Treatises on the Arts of Painting*, 2 vols. (1849; reprint, New York, 1967), 1:166–203 for the Latin text and English translation, and 182–83 for the citation.

[56]John G. Hawthorne and Cyril Stanley Smith, *Theophilus: The Various Arts* (1963; reprint, New York, 1979), pp. 11–13. See also C. R. Dodwell, ed. and trans., *Theophilus: The Various Arts* (London, 1961). For a valuable discussion of Theophilus and his view of the labor of craftsmen, see George Ovitt, Jr., *The Restoration of Perfection: Labor and Technology in Medieval Culture* (New Brunswick, N.J., 1987), pp. 168–73. See also Gaines Post, Kimon Giocarinis, and Richard Kay, "The Medieval Heritage of a Humanistic Ideal: '*Scientia donum Dei est. Unde vendi non potest,*' " *Traditio* 11 (1955): 195–234, a

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Having abrogated credit for his own authorship through pseudonymousness, Theophilus advocated the open sharing of human skill and knowledge for the dual purposes of human advancement and the glory of God. Yet he lived in a time when technological developments in agriculture and in the harnessing of power, among other factors, had led to population growth, the development of commerce, and the rise of urbanism.[57] The early stages of this "technical revolution" took place silently. For such medieval innovations as the adoption of the stirrup, the improvement of the horse's harness, the development of the three-field system of cultivation, and the expansion of waterpower, I know of no evidence for an interest in authorship, credit to inventors, or the appropriation of technical knowledge. Such lack of contemporary comment on the "metahistory" of early medieval technical invention is often attributed to the widespread illiteracy of practitioners. Yet some of these technical innovations occurred in monastic and other environments where literacy did exist. A second explanation seems more probable, namely, that for numerous medieval practitioners, issues involving credit, invention, and authorship were of no concern, even if the relevant concepts had been available. The rewards for improvements in agricultural, military, and power technologies consisted of more food, greater safety, and relief from backbreaking labor. Increasingly those rewards also included the ability to possess luxuries, of great symbolic value to the dominant classes.

"Intellectual property" attitudes did not develop as part of the technological advances that contributed to the growth of urbanism. I would propose, rather, that such an outlook emerged from the context of medieval urbanism itself. Medieval cities and the market economies that developed within and among them provided the essential context for the emergence of a fully developed concept of

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study of the medieval ideal that knowledge was freely given by God and should not therefore be sold.

[57]Initial bibliography for the complex issues of labor and technology in the medieval period includes Marc Bloch, *Land and Work in Mediaeval Europe: Selected Papers by Marc Bloch,* trans. J. E. Anderson (London, 1967), and his "Les 'inventions' médiévales," *Annales d'histoire économique et sociale* 7 (1935): 634–43; Jean Gimpel, *The Medieval Machine: The Industrial Revolution of the Middle Ages* (New York, 1976); Jacques Le Goff, *Time, Work, and Culture in the Middle Ages,* trans. Arthur Goldhammer (Chicago, 1980); Ovitt (n. 56 above); Brian Stock, "Science, Technology, and Economic Progress in the Early Middle Ages," in *Science in the Middle Ages,* ed. David C. Lindberg (Chicago, 1978), pp. 1–51; and Lynn White, jr., *Medieval Technology and Social Change* (Oxford, 1962), and his *Medieval Religion and Technology: Collected Essays* (Berkeley and Los Angeles, 1978).

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intellectual property. That concept is first evident within the regulations of the craft guilds into which many urban artisans had been organized (or had organized themselves) by the mid-13th century.

Medieval Craft Guilds Fostered Proprietary Attitudes toward (Intangible) Craft Knowledge

Nearly all medieval craft guilds were urban, but beyond that there was great diversity among them. Variations were the result of differences in geographic, social, and linguistic contexts and also depended on the nature of the craft or crafts included within the guild and the relationship of the guild to the polity. The guild regulations or capitularies that were recorded in various urban centers around the mid-13th century, as well as the other archival materials that surround them, are unique documents that have been relatively neglected by historians of technology and of cultural and intellectual history.[58]

The significance of the medieval craft guilds includes their role in the development of proprietary attitudes toward craft knowledge. In promoting attitudes of ownership toward intangible property—craft knowledge and processes as distinct from material products—the guilds developed the concept of "intellectual property" without ever calling it that. The view that craft knowledge was intangible property with commercial value developed in this context quite apart from notions of individual authorship. Neither was it necessarily tied to innovation. "Intellectual property" became an aspect of corporate ownership—whether the corporation was more or less autonomous, or was closely tied to the government of the commune.

This thesis can be illustrated with the example of glassmakers of Venice, famous for making the finest glass in Europe and also for the secrecy of their techniques. Venetian guilds, as Richard Mackenney has expressed it, "fulfilled a dual function as administrative units for the government and as associations of craftsmen." The glassmakers, called *fieroli,* were required to move to the nearby island of Murano in 1291 because of the fire hazard posed by their furnaces. Their

[58]The best introduction to medieval European craft guilds is Sylvia L. Thrupp, "The Gilds," in *The Cambridge Economic History of Europe,* vol. 3, *Economic Organization and Policies in the Middle Ages* (Cambridge, 1963), pp. 230–80 and 624–34. For the relationship of the guilds to the development of political thought, see Antony Black, *Guilds and Civil Society in European Political Thought from the Twelfth Century to the Present* (Ithaca, N.Y., 1984). See also Bo Gustafsson, "The Rise and Economic Behaviour of Medieval Craft Guilds: An Economic-Theoretical Interpretation," *Scandinavian Economic History Review* 35 (1987): 1–40, who emphasizes quality control as central to the origin and function of the guilds.

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corporate organization, as was true of the other Venetian guilds, included the guild itself (*arte*) and an associated religious confraternity (*scuola*). The guild was governed by the *Giustizia Vecchia,* a magistracy chosen by and representative of the commune of Venice. More direct oversight was provided by a *gastaldus* and other officers, *judices,* who were elected by guild members but beholden to the *Giustizia Vecchia* and to the regulations approved by it. Thus Venetian glassmakers, like other Venetian artisans, were closely controlled by the commune.[59]

The glassmakers' regulations (which constitute the extant capitulary) concerned governance of guild members and the regulation of the guild itself. Glassworkers included furnace owners (*padroni di fornace*), masters, and workers—some of them apprentices, others, low-level laborers with no chance of becoming guild members.[60] Glassmaking in Venice was restricted to guild members who had taken the oath. They were required to join the *scuola* of the guild and to hear the regulations read once per year.[61] Guild regulations pre-

[59]Richard Mackenney, *Tradesmen and Traders: The World of the Guilds in Venice and Europe, c.1250–c.1650* (Totowa, N.J., 1987), p. 9 (for citation) and pp. 1–43 for the relationship of the Venetian guilds to the government; and see also Dennis Romano, *Patricians and Popolani: The Social Foundations of the Venetian Renaissance State* (Baltimore, 1987), pp. 65–90. Fundamental to the study of Venetian guilds is the work of Giovanni Monticolo. See G. Monticolo and E. Besta, *I capitolari delle Arti Veneziane sottoposte alla Giustizia e poi alla Giustizia Vecchia dalle origini al MCCCXXX,* 3 vols., Fonti per la storia d'Italia, 26–28 (Rome, 1896–1914), which includes important introductory material as well as the guild capitularies themselves. For a study of the confraternities, see Lia Sbriziolo, "Per la storia delle confraternite veneziane: Dalle Deliberazioni miste (1310–1476) del consiglio dei dieci, *scolai comunes,* artigiane e nazionali," *Atti dell'Istituto Veneto di Scienze, Lettere ed Arti: Classe di scienze morali, lettere ed arti* 126 (1967–68): 405–42. For the *Giustizia Vecchia,* see G. Monticolo, *L'Ufficio della Giustizia Vecchia a Venezia dalle origini sino al 1330* (Venice, 1892).

[60]For the capitulary of the glassmakers (*fieroli*), see Monticolo and Besta, 2:61–98. The *fieroli* are to be distinguished from the *cristollari* who worked in rock crystal. For useful discussions of the Venetian glassmakers and their art, see esp. Franco Brunello, *Arti e mestieri a Venezia nel medioevo e nel rinascimento* (Vicenza, 1980), pp. 17–30; Bartolomeo Cecchetti, "Delle origini e dello svolgimento dell'arte vetraria muranese," *Atti dell'Istituto Veneto,* ser. 4, 1 (1871–72): 1681–1798; Bartolomeo Cecchetti and Vincenzo Zanetti, *Monografia della Vetraria Veneziana e Muranese* (Venice, 1874); Astone Gasparetto, *Il vetro di Murano dalle origini ad oggi* (Venice, 1958); Giovanni Mariacher, *L'arte del vetro* (Verona, 1954); and Mariacher, *I vetri di Murano* (Milan, 1967). An important article that discusses recent technical and archival research is Rosa Barovier Mentasti, "Tecnica del vetro nella Venezia del cinquecento," in *Cultura, scienze e techniche nella Venezia del cinquecento,* Atti del Convegno Internazionale di Studio Giovan Battista Benedetti e il suo tempo (Venice, 1987), pp. 473–82. For a convenient summary, Frederic C. Lane, *Venice: A Maritime Republic* (Baltimore, 1973), pp. 156–60.

[61]Monticolo and Besta, 2:64, chap. II; 2:67, chap. XI; and 2:70, chap. XXIII.

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scribed such matters as legal work days for glassmaking,[62] instructions for the election of guild officials,[63] judicial procedures,[64] rules governing apprenticeships, and the relationship of masters and *padroni*.[65] Handling stolen goods and selling defective glass products or non-Venetian glass products were forbidden.[66] Only a few regulations dealt with the actual technical processes of glassmaking, although the type of wood to be used in the furnaces was specified, as was the number of openings (*boche*) allowed in the main furnace, both of which factors affected the quality of the product.[67]

Yet glassmaking was a complex process that required three furnaces (sometimes combined into one or two structures). The clarity,

[62]Ibid., 2:62–64, chap. I, 2:85, chap. LXXII, and 2:86, chap. LXXV, designate the feast days that the *fieroli* must celebrate, whereas 2:89–91, chaps. LXXXI and LXXXII, and 2:96–97, chap. LXXXXIII, allude to the seven-month work and five-month selling periods to which the glassworkers were compelled to conform.

[63]Two of the most important election rules prescribed that the *gastaldus* could hold office for only one year and forbade *padroni* of the furnaces to run for office in either the *arte* or the *scuola:* ibid., 2:75, chap. XXXXI (November 1265), and 2:76–77, chap. XXXXV, respectively.

[64]Adjudication of most disputes, as well as enforcement of the regulations, were the responsibilities of the *gastaldus* and the *judices*. Many chapters of the capitulary concern these functions. They prescribe accountability between *gastaldus* and *judices;* insist that adjudication occur when requested; prevent interference of the *gastaldus* in the cases of the *judices;* instruct that the *judices* are solicitous both of the needs and utility of the guild and the good of Venetians; order that the *judices* watch the *gastaldus,* council him if he does not follow the regulations, and if necessary report him to higher authorities; fix the limit of fines appropriate for the judgments of the *gastaldus* and *judices;* and fine the *gastaldus* who does not open his office for justice. The regulations insure that lesser disputes and infractions are adjudicated, that control over the officers of the guild is maintained, and that the guild regulations are followed. See ibid., 2:67–68, chaps. XII, XIII, XIIII; 2:68–70, chaps. XVII and XVIII, XVIIII, XX, XXI, XXII; 2:72, chap. XXX and 83, chap. LXIII; and 2:92, chap. LXXXIIII.

[65]Ibid., 2:65, chap. VI; 2:71, chaps. XXIIII and XXV; 2:74, chap. XXXVIII; 2:82, chap. LVII; 2:83, chaps. LXI, LXII, and LXIIII; and 2:85, chaps. LXVIIII, LXX. The regulations specify the minimum age of apprentices (eight years) and include measures to reduce competition among glassworkers, such as the prohibition against persuading another's apprentice to work for oneself.

[66]Ibid., 2:67–68, chap. XV, and 2:79, chap. LII (concerning stolen goods); 2:73, chap. XXXIIII (concerning the sale of imperfect products); 2:86, chap. LXXIII (concerning the selling of non-Venetian glass in Venice).

[67]The wood to be used in the furnace was specified (first as alder or willow, later as alder only). Not more than three or four *boche* (mouths) were allowed in the main furnace. Both of these factors affected the clarity of the glass. Ibid., 2:65, chap. VII, 2:94–95, chaps. LXXXVIII and LXXXX (concerning wood), 2:65, chap. V, 2:91–92, chap. LXXXIII, and 2:92–93, chap. LXXXVI (concerning furnace "mouths"). See also 2:95–96, chaps. LXXXXI and LXXXXII for attempts to regulate the use of *fulginus,* the potassium carbonate that northern glassmakers used instead of the sodium carbonate of Venetian glass (see Gasparetto [n. 60 above], pp. 239 and 247).

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strength, and quality of the final product depended on the nature, purity, and mixture of the original ingredients, on a detailed knowledge of the proportions and ways of combining them, and on the effect of furnace temperatures on the materials at various stages of the process. Requisite knowledge of coloration formulas and techniques and of glassblowing added to the specialized nature of the craft. Venetian glass was superior to the glasses of northern Europe in part because the Venetians had access to superior materials, including Syrian soda ash (sodium carbonate), made by burning seaweed that provided ballast for Venetian ships from the East. Soda ash produced a clearer glass than the potash (potassium carbonate obtained from burning hardwoods) used by northern glassmakers. Yet the skill and knowledge of Venetian glassmakers were also intrinsic to their success. And as the fame of Venetian glassware increased, the special knowledge of Venetian glassmakers acquired ever greater value.[68]

The commune of Venice considered that the craft knowledge of the glassmakers of Murano was communal property to be used for the benefit of Venice and the guild. By means of lucrative sales, Venetian glass products spread across Europe, but the export of the craft itself was strictly forbidden. Glassworkers were prohibited from plying their craft outside of Venice. Guild capitularies of 1271 specified that "anyone of the aforementioned art who will have gone outside Venice for the occasion of practicing the said art" will pay a fine, and that further the "*gastaldus* must not accept the oath [of guild membership] from men who will have gone out beyond Venice with the reason of this art without the permission of the justices."[69] In 1281 the regulation was repeated with higher fines levied as penalties.[70] In 1295 the *gastaldus* and other officials of the guild petitioned the council to punish glassworkers who worked outside by banishing them from the guild and from working the art in Venice for a prescribed time. They complained that the small fine was ineffective and that, in a short time, furnaces had appeared in Vicenza, Padua, Mantua, Ferrara, Ravenna, Ancona, and Bononia. The petition also requested that documents necessary for exporting glassworking materials (e.g., sand

[68]For glassmaking technology in both northern and southern Europe in the late medieval period, see R. J. Charleston and L. M. Angus-Butterworth, "Glass," in *A History of Technology*, vol. 3, *From the Renaissance to the Industrial Revolution, c.1500–c.1750*, ed. Charles Singer et al. (New York and London, 1957), pp. 206–44. For Venetian glassmaking, see n. 60 above.

[69]Monticolo and Besta, 2:66, chap. VIII, "unusquisque de arte predicta qui exierit extra Venecias occassione exercendi dictam artem, . . ."; "gastaldus non debeat accipere sacramentum ab hominibus qui istius artis causa exierit extra Venecias sine licencia iusticiariorum" (fine = 10 *denariorum venecialium*).

[70]Monticolo and Besta, 2:79, chap. LI. The fine was raised to 5 *libras*.

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and alum) be withheld from such men, that these men "may be forsaken and moreover may be disparaged," and that guild members be required to inform officials about members that they knew of or discovered working outside the city. The petition was granted.[71]

Yet effective control was elusive. In 1303, guild capitularies note that the law was shouted out in San Marco and the Rialto: any glassworker who had worked outside of Venice must come before the Lords Justice to make their excuse and may be banned perpetually from the guild.[72] In 1313 it is recorded that a glassworker, Petrus Caldera, was indeed banned from the guild for working outside the city and could only be restored to grace after paying a fine of 200 libras and promising not to repeat the infraction. His two glassmaker sons-in-law posted the deposit for him, and it was noted that if he worked outside again he must pay the fine again, and that they also would be fined. The strain on family relationships must have been great, for in 1315 one son-in-law, Donatus, renounced his security pledge.[73] In the same year, the *Giustizia Vecchia* tried another tactic. It had a crier shout in the Rialto and San Marco that all glassworkers working outside of Venice must return within two months and could do so without penalty. Those who stayed away would be subject to the penalties outlined above.[74]

The story of a 15th-century glassmaker's career illustrates the importance of craft secrets—the essential intellectual property of the Venetian glass industry. One Giorgio Ballarin was an employee of a long-established glassmaking family, the Barovier. One day in their temporary absence he broke into the place where they kept their secret formulas, stole them, and brought them to a rival glassmaker. He thereby was able to marry the glassmaker's daughter and acquire a furnace of his own.[75] Even among glassmakers in Venice itself, secret knowledge of craft processes provided significant advantages.

This understanding of craft processes as intangible property with significant commercial value was an important development in intellectual history that led to the emergence of patents. The Venetian state, Venetian glassmakers disappearing over the border, and those

[71]Ibid., 2:88–89, chap. LXXX, "desertantur et extrinseci elevantur."

[72]Ibid., 2:92, chap. LXXXV.

[73]Ibid., 2:97, chap. LXXXXIIII.

[74]Ibid., 2:97–98, chap. LXXXXV.

[75]Lane (n. 60 above), p. 159. For documentation of the incident, see Emmanuele Antonio Cicogna, *Delle inscrizioni veneziane* (1853; reprint, Bologna, 1970), vol. 6, pt. 1, pp. 466–71. Mentasti (n. 60 above) elaborates the increasingly complex technology of 15th-century Venetian glassmaking, the secrets of which are still in the process of being discovered.

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who stayed at home all recognized that knowledge of glassmaking processes was a precious commodity. Whereas both Venice itself and individual Venetian glassmakers were significant players in this development, no exclusive claim for them is made. In the high and late Middle Ages, many urban centers and the guilds associated with them claimed proprietary rights over certain crafts and industries. Those claims at times included "intellectual property," especially in highly technical, complex crafts. This is an area in which far more research is needed. Given the diversity of medieval cities, guilds, and crafts, a uniform picture should not be anticipated. Yet it is clear that within medieval cities the attitude developed that craft processes constituted intangible property with commercial value subject to conditions of ownership. It is also clear that Italian cities, and particularly Venice, were centrally important to the development of such attitudes.

Early Patents and Patent Laws: A Product of Medieval Urbanism

One result of the view that craft knowledge constituted a form of intangible property was the development of the patent. Patents and patent laws emerged as an aspect of medieval urban economic policies associated with maintaining control over the crafts, maintaining possession of the benefits of craft knowledge, and encouraging innovation. The latter was achieved by both rewarding local invention and compensating strangers who brought new techniques and processes into the cities from other localities. There is evidence of a few patents in the 13th century. By the 15th there is a steady flow of patent awards in Venice. The patent was a limited monopoly awarded for the possession of locally novel craft knowledge or for the invention of new devices.[76]

[76]The best general introduction to early European patents is Marcel Silberstein, *Erfindungsschutz und merkantilistische Gewerbeprivilegien* (Zurich, 1961). For Venetian patents, see Giulio Mandich, "Le privative industriali veneziane (1450–1550)," *Rivista del diritto commerciale e del deritto generale delle obbligazioni* 34 (September–October 1936): 511–47, and his "Primi riconoscimenti veneziani di un diritto di privativa agli inventori," *Rivista di diritto industriale* 7 (1958): 101–55. The first has been translated by Frank Prager—Giulio Mandich, "Venetian Patents (1450–1550)," *Journal of the Patent Office Society* 30 (March 1948): 166–224. Prager's "translation" of the second—Giulio Mandich, "Venetian Origins of Inventors' Rights," *Journal of the Patent Office Society* 42 (June 1960): 378–82—involves the abridgement of fifty-five pages to five and should not be substituted for the original. Until recently, the early history of patents in continental Europe had been investigated primarily by patent lawyers interested in the antecedents of their own discipline. The great value of this work was that it pointed out the existence and location of early patents. Yet much of it is marred by inadequate documentation, overgeneralization, and an anecdotal quality that fails to explore patents within the context of economic history and the history of technology. General discussions include Maximilian Frumkin, "The Origin of Patents," *Journal of the Patent*

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A law passed in 1297 by the Great Council of Venice illustrates the principle of urban regulation combined with communal encouragement of innovation. The Venetian law concerned the manufacture and sale of medicines. They could be sold only in shops organized as public firms that were subject to the strict supervision and control of the *Giustizia.* Yet the invention of new medicines was encouraged by the following provision: "And if any physician wishes to make any of his own medicine in secret, he may be empowered to make it, if only, of course, of the best materials, and all may hold in confidence, and all guild members may swear not to interject themselves into the above mentioned [matter]."[77] In effect, the council gave the physician

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Office Society 27 (March 1945): 143–49, "Early History of Patents for Invention," *Transactions of the Newcomen Society* 26 (1947–49): 47–56, and "The Early History of Patents for Invention," *Transactions of the Chartered Institute of Patent Agents* 66 (1947–48): 20–69; Frank D. Prager, "The Early Growth and Influence of Intellectual Property," *Journal of the Patent Office Society* 34 (February 1952): 106–40, "Examination of Inventions from the Middle Ages to 1836," *Journal of the Patent Office Society* 46 (April 1964): 268–91, and "A History of Intellectual Property from 1545 to 1787," *Journal of the Patent Office Society* 26 (November 1944): 711–60. For 16th-century German patents, see Rolf-Jürgen Gleitsmann, " 'Wir wissen aber, Gott Lob, was wir Thuen': Erfinderprivilegien und Technologischer Wandel im 16. Jahrhundert," *Zeitschrift für Unternehmensgeschichte* 30 (1985): 69–95; Hansjörg Pohlmann, "Neue Materialien zur Frühentwicklung des deutschen Erfinderschutzes im 16. Jahrhundert," *Gewerblicher Rechtsschutz und Urheberrecht* 62 (June 1960): 272–83, which includes extensive documentation. (The "translation" by Frank D. Prager, "The Inventor's Right in Early German Law: Materials of the Time from 1531 to 1700," *Journal of the Patent Office Society* 43 [February 1961]: 121–39, involves the elimination of more than half the footnotes, among other abridgements.) See, finally, Silberstein, esp. pp. 41–90. For French patents, see Maximilian Frumkin, "Les anciens brevets d'invention: Les pays du continent européen au XVII[e] siècle," *Archives internationales d'histoire des sciences* 7 (July–December 1954): 315–23; and Jacques Isoré, "De l'existence des brevets d'invention en droit français avant 1791," *Revue historique de droit français et étranger,* 4 ser. 16 (1937): 94–130. For the Netherlands, see Gerard Doorman, *Patents for Invention in the Netherlands during the 16th, 17th, and 18th centuries,* trans. Johan Meijer (The Hague, 1942). English patents and invention are the subject of an exemplary study, Christine MacLeod, *Inventing the Industrial Revolution: The English Patent System, 1660–1800* (Cambridge, 1988), which refers to prior bibliography on English patents. A perceptive article that focuses on both books and inventions and the development of legal concepts around them is Remo Franceschelli, "Le origini e lo svolgimento del diritto industriale nei primi secoli dell'arte della stampa," *Rivista di diritto industriale* 1, pt. 1 (1952): 151–95.

[77]Mandich, "Primi riconoscimenti veneziani," pp. 105–6. The document—Archivio stato veneziana, Maggior Consiglio, Deliberazioni, Liber Pilosus—is published by R. Cessi, *Deliberazioni del Maggior Consiglio di Venezia* (Bologna, 1934), 3:423—"et si aliquis medicus vellet facere aliquam medicinam suam secreto, teneatur eam facere modo scilicet de melioribus rebus, et teneant omnis in credentia, et jurent omnes stationarii non intromittere se de predictis."

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monopoly rights over his own invention by allowing him to protect his secret—and sell it—as long as he used high-quality ingredients.

Venetian interest in actively promoting the economy is evidenced by 13th-century licenses. The historian of Venetian patents, Giulo Mandich, pointed to a number of documents of the years 1281–96 in which the Grand Council gave licenses for building various kinds of mills. One in the year 1281 seems to refer to a new invention involving a windmill, although the language is imprecise. In 1323 the council agreed to support one Joannes Teuthonicus "inzenerius molendinorum" as he made machines sufficient for the needs of Venice. It would support the cost of the "experientia" up to 80 ducats. Then, the *Doge* and *Signoria* with the help of two experts would establish a just compensation for the inventor (or importer) of the new mills "if said work is good and useful for the commune." In the same century licenses were given for new dredging machines.[78]

Patents and patent law came out of this interest in promoting the economy of the city and encouraging useful inventions. Yet true patents—awards of monopolies of specific duration for innovations—are far more common in the 15th century than before that time. Thus, in 1416, one Franciscus Petri, from the island of Rhodes, petitioned the Council of Venice for a patent for "structures with pestles for fulling fabrics." It was granted "because they full thus perfectly, better than the fullers existing in the waters surrounding" the Rialto. For fifty years, no one but Petri and his heirs were permitted to build or have built similar fulling devices, nor could any additions or reconstructions be made of any part other than what he provided. For, the council emphasized, "his fulling is better than that of the usual devices and different therefrom and his device is superior to the usual fulling devices and better than the same."[79] It is notable that Venice here awarded a patent to a foreigner (against Venetian fullers) for a superior machine. Original authorship was not an issue. Petri was awarded the patent because he possessed the new invention. Whether he invented it himself or got it from Rhodes is unknown. Petri's patent was not an isolated case. Numerous patents were issued for various devices from that time forward.[80]

[78]Mandich, "Primi riconoscimenti veneziani," pp. 106–10, 107 for material on the windmill.

[79]Ibid., pp. 115–16 and pp. 149–50 (for publication of the document). For an English translation of the entire document, see Mandich, "Venetian Origins," trans. Prager (n. 76 above), p. 379, n. 6. I have followed Prager's translation, although with minor changes.

[80]See Mandich, "Primi riconoscimenti veneziani," pp. 116–55, for a detailed discussion of patents and related provisions after Petri's fulling machine patent to 1547.

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The Venetian Senate passed the first general patent law (as distinct from individual grants of patent) on March 19, 1474. The law elaborated that the greatness of the city of Venice had drawn men of diverse origins who possessed "very sharp ingenuity" and were "skilled in devising and discovering various ingenious devices." Provided that others did not increase their own honor by possessing these works, these men would exercise their ingenuity and discover and make things useful and beneficial to the state. Thus the law required that each person who invents a new and ingenious device, not made before in the Venetian domain, must give notice of it at the *Provveditori di Comun.* No one within the Venetian domain could make a similar device for ten years. If anyone violated the law, the "author and inventor" could report it to city officials. The infringer could be required to pay 100 ducats and the device might immediately be destroyed. However, the Venetian government could itself take and use it for its own needs as long as others did not.[81] This law did not embody a "new" principle, but was a codification of prior practice. It was followed by a steady stream of Venetian patent awards from the late 15th through the 16th centuries.[82]

Patents for new inventions were awarded in other localities as well—for example, the 1421 patent that the Florentine commune awarded to Brunelleschi for a new ship he had invented to haul loads on the Arno River. Brunelleschi's petition for the patent described him as "a man of the most perspicacious intellect, industry and invention." The petition claimed that the newly invented ship could haul loads more cheaply and that it would provide further (unspecified) benefits to merchants and others. However, Brunelleschi "refuses to make such machine available to the public in order that the fruit of his genius and skill may not be reaped by another without his will and consent, and that, if he enjoyed some prerogative concerning this, he would open up what he is hiding and would disclose it to all." A patent would allow the matter to be brought to light for the benefit of Brunelleschi, the whole country, and others. Additionally, Brunelleschi would be animated more

[81]For the text of the law and a discussion, see Mandich, "Le privative industriali" (n. 76 above), pp. 518–19, "accutissimi ingegni"; "apti ad excogitar et trovar varij ingegniosi artificij"; "auctor et inventor." For English translations and further discussions, see Frumkin, "Early History of Patents" (n. 76 above), p. 49; Mandich, "Venetian Patents," trans. Prager (n. 76 above), pp. 176–77; and Jeremy Phillips, "The English Patent as a Reward for Invention: The Importation of an Idea," *Journal of Legal History* 3 (May 1982): 75–76.

[82]Mandich, "Le privative industriali" (n. 76 above), pp. 537–47, lists 109 Venetian patents awarded between 1475 and 1549.

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fervently "to even higher pursuits, and stimulated to more subtle investigations."[83] The justification for the patent on the basis of Brunelleschi's own authorship and inventive abilities and the broad nature of the rights granted are notable.

Once an inventor holds the view that his inventions constitute intellectual property, two forms of protection may be available. The first is the limited monopoly or patent. The second is secrecy. Brunelleschi seems to have tried both. His views on secrecy were recorded by the Sienese author of books on machines, Mariano Taccola. As recorded by Taccola, Brunelleschi admonished:

> Do not share your inventions with many, share them only with a few who understand and love the sciences. To disclose too much of one's inventions and achievements is one and the same thing as to give up the fruit of one's ingenuity. Many are ready when listening to the inventor, to belittle and deny his achievements, so that he will no longer be heard in honorable places, but after some months or a year they use the inventor's words, in speech or writing or design. They boldly call themselves the inventors of the things that they first condemned, and attribute the glory of another to themselves.[84]

Craft secrecy was an important form of protection in the late medieval/early modern period. Yet the use of patents became increasingly common in the 16th century. The Italian glassmakers, aware of the value of their intangible property, were instrumental in obtaining patents beyond Italy. In so doing they exported not only Venetian glassmaking techniques (often their patents required them to train

[83]Frank D. Prager, "Brunelleschi's Patent," *Journal of the Patent Office Society* 28 (February 1946): 109–35 and 109–10 for a translation of the petition and grant. The document was published by Giovanni Gaye, *Carteggio Inedito d'Artisti dei Secoli XIV, XV, XVI,* 3 vols. (Florence, 1839), 1:547–49. The council of Florence granted a three-year patent specifying that no one except Brunelleschi could have or use on Florentine waters any new ship or machine designed for transporting goods on water. Only "familiar and usual" ships could be used by others. Any "new or newly shaped machine would be burned." All new ships must be made either by Brunelleschi or "with his will and consent." All goods transported in the newly invented ship would be free of levies and taxes not previously imposed for three years. Prager elaborates the evidence that Brunelleschi's invention was a technical fiasco that failed to carry a single load to Florence. See also Frank D. Prager and Gustina Scaglia, *Brunelleschi: Studies of His Technology and Inventions* (Cambridge, Mass., 1970), pp. 111–23.

[84]Frank D. Prager, "A Manuscript of Taccola, Quoting Brunelleschi, on Problems of Inventors and Builders," *Proceedings of the American Philosophical Society* 112 (June 1968): 131–49; and Frank D. Prager and Gustina Scaglia, *Mariano Taccola and His Book "De ingeneis"* (Cambridge, Mass., 1972), pp. 11–13.

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local apprentices) but also the Italian notion of the patent as a way of promoting economic growth and development.[85]

Italians other than glassworkers also aided the spread of the use of patents. In England, for example, Jacobus Acontius petitioned Elizabeth I for a patent with an explicit justification that a patent grant should be an award for inventors. His petition reads in part: "Nothing is more honest than that those who by searching have found out things useful to the public should have some fruit of their rights and labours, as meanwhile they abandon all other modes of gain, are at much expense in experiments, and often sustain much loss, as has happened to me." He claimed to have discovered "new kinds of wheel machines, and of furnaces for dyers and brewers." He lamented that if they were known they would be used without his consent (unless there was a penalty), while he, "poor with expenses and labour, shall have no returns." The grant of the petition noted that "it is right that inventors should be rewarded and protected against others making profits out of their discoveries." Jeremy Phillips has argued convincingly that the Venetian patent law of 1474 exerted a significant influence on this petition.[86]

European and English patents in the 15th and 16th centuries consisted of limited monopolies for novel machines and processes, sometimes awarded on the basis of authorship, at other times on the basis of possession of knowledge about particular processes or mechanisms. Patents for new inventions and processes constituted one of several methods whereby governments encouraged economic development. Patents were associated with other kinds of rewards for novel and useful inventions (such as tax exemptions) and were also associated with patents and privileges awarded on the basis of court favor, or for other reasons. Patent law has also been associated with laws awarding mineral, land, and water rights.[87] Historians of patents for

[85]Early patents were issued to Italian glassmakers in Antwerp, Holland, England, France, Germany, and Austria. See Frumkin, "Early History" (n. 76 above), pp. 50–54. For a detailed study of Venetian glassmaking in the Netherlands from the 16th century, see H. Schuermans, "Verres 'façon de Venise' fabriqués aux Pays-Bas," *Bulletin des commissions royales d'art et d'archéologie* 27 (1888): 197–301. As Christine MacLeod has demonstrated in "Accident or Design? George Ravenscroft's Patent and the Invention of Lead-Crystal Glass," *Technology and Culture* 28 (October 1987): 776–803, Venetian glassmaking was influencing the patent system well into the 17th century.

[86]Phillips (n. 81 above), pp. 71–79, esp. p. 71. For Acontius's career, see Lynn White, jr., "Jacopo Aconcio as an Engineer," in *Medieval Religion and Technology* (n. 57 above), pp. 149–73.

[87]See Mandich,"Primi riconoscimenti veneziani" (n. 76 above), esp. pp. 101–14; and Giulio Mandich, "Privilegi minerari e agricoli a Venezia nel sec. XV.," *Rivista di diritto industriale* 7 (1958): 327–59.

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invention have been concerned to distinguish "true patents" from these other types of awards. Although such distinctions are necessary and legitimate, patents for new inventions should also be seen in context, as one of several ways by which late-medieval communes and rulers promoted their economic and sometimes political aims.

Intellectual property values emerged from a guild/commune context but were often disseminated by individuals breaking away from the guild. The ingenuity of individuals and the importance of authorship would have been enhanced by such a process. For example, Brunelleschi's patent claims his own ingenuity and invention as justification. His refusal to work within guild restrictions and his conflicts with the Florentine guilds are well known.[88] Early Venetian patents often awarded monopolies to outsiders in ways that cannot have been favorable to native guildsmen.[89] It might be said that the guild/commune environment encouraged the concept and value of intellectual property, whereas individuals walking away from that context carried these new views with them but applied them to different interests. They claimed possession of craft knowledge and the authorship of their own inventions as both deserving of patent awards. Frequently, they added two additional justifications—the utility of the invention and the good of the community.

The Idea of Genius and the Emergence of the Priority Dispute

In conclusion, it is possible only to suggest (without full elaboration) two related developments in intellectual history that had an important influence on the history of intellectual property. The first concerns the belief in the importance of individual ingenuity and genius in creation and invention. The notion of individual genius developed significantly in the 15th century and became increasingly important in the 16th. The second involves the emergence of the priority dispute, which was a direct result of this enhanced view of individual genius as intrinsic to discovery and invention.

In an illuminating study, Martin Kemp has pointed to the 15th-century development of the notion of inventive imagination in the visual arts—a concept that was to be elaborated more extensively in the 16th century. Kemp has identified three 15th-century individuals—Brunelleschi, Leonardo da Vinci, and Francesco di Giorgio—who were singled out by their own contemporaries for their

[88]Prager, "Brunelleschi's Patent" (n. 83 above), p. 117.

[89]See text above for the example of the patent to Petri from Rhodes for his new fulling machine.

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creative ingenuity.[90] Perhaps it is not coincidental that all three were inventors of machines and devices as well as practitioners in the visual arts.

Like their predecessor Brunelleschi, Leonardo da Vinci and Francesco di Giorgio were also concerned with protecting the products of their own ingenuity. Leonardo's mirror writing may have been the natural product of a left-handed autodidact, but it also protected his ideas and inventions from facile borrowings.[91] Francesco di Giorgio lamented the hardships he endured as an inventor, admitting that he had often been tempted not "to wish to manifest any of my machines for the reason that I have acquired the knowledge of that with my great cost of experience and grave inconvenience." He had been rewarded with ingratitude and with the incomprehension of those who failed to understand the long labors of invention. Yet these things were nothing compared to the "great inconvenience and disturbance to the soul" that occurs when the ignorant glory themselves with the inventions of others.[92]

The notion of individual ingenuity gained in importance when creativity itself was conceived as the production of novelty rather than as an imitative activity. A significant text is Nicholas of Cusa's mid-15th-century dialogue, the *Idiota,* wherein the maker of pots and spoons is described as superior to the painter. The painter imitates nature, but the maker of pots creates something with no model in the natural world. Thus the potmaker's work is closer to that of the divine creator than is the work of the painter.[93] Yet in art theory itself, the

[90]Martin Kemp, "From 'Mimesis' to 'Fantasia': The Quattrocento Vocabulary of Creation, Inspiration and Genius in the Visual Arts," *Viator* 8 (1977): 347–98; see also Kemp, "The 'Super-artist' as Genius: The Sixteenth-Century View," and (for the ancient background) Penelope Murray, "Poetic Genius and Its Classical Origins," both in *Genius: The History of an Idea,* ed. Penelope Murray (New York, 1989), pp. 32–53 and 9–31, respectively. An important early study on the idea of genius is Edgar Zilsel, *Die Entstehung des Geniebegriffes: Ein Beitrag zur Ideengeschichte der Antike und des Frühkapitalismus* (Tübingen, 1926).

[91]Martin Kemp, *Leonardo da Vinci: The Marvellous Works of Nature and Man* (Cambridge, Mass., 1981), is the best introduction.

[92]Francesco di Giorgio Martini, *Trattati di architettura ingegneria e arte militare,* ed. Corrado Maltese, transcription Livia Maltese Degrassi, 2 vols. (Milan, 1967), 2:492, "di non volere manifestare alcuna mia macchina, peroché avendo io acquistata la notizia di quelle con grande mia spesa di esperienzia e grave incomodo"; "maggiore incomodo all'animo e molestia."

[93]Nicholas of Cusa, *Opera omnia,* ed. L. Bauer (Leipzig, 1937), 5:95–98 (*Idiota de mente,* 2.62–64). There is an anonymous English translation—*The Idiot in Four Books* (London, 1650). For the importance of this dialogue for the idea of creativity, see Hans Blumenberg, " 'Nachahmung der Natur': Zur Vorgeschichte der Idee des schöpferischen Menschen," *Studium Generale* 10 (1957): 266–83.

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notion of imitation as a descriptive concept gave way to the 16th-century concept of "fantasia," an inspired form of creativity sparked by the genius of the painter. Mannerist art theory gave high priority to the individual "manner" of the painter. For the first time, as well, serious analysis was undertaken of individual painting styles that result from the individuality of painters.[94]

When the notion of individual ingenuity was joined to the concept of intellectual property, one result was the priority dispute. As Robert Merton observed, disagreements among contemporaries concerning who first invented or discovered something seem to have appeared initially in the 16th century.[95] Early-modern priority disputes were bitterly fought and frequently involved accusations of theft or plagiarism.[96] Further investigation should clarify areas of contention more

[94]For an analysis of "imitation" from rhetoric to art theory, see Rensselaer W. Lee, *'Ut pictura poesis': The Humanistic Theory of Painting* (New York, 1967); John R. Spencer, " 'Ut rhetorica pictura': A Study in Quattrocento Theory of Painting," *Journal of the Warburg and Courtauld Institutes* 20 (1957): 26–44. For mannerism and mannerist art theory, see Arnold Hauser, *Mannerism: The Crisis of the Renaissance and the Origin of Modern Art,* trans. Eric Mosbacher, 2 vols. (London, 1965); and Denis Mahon, *Studies in Seicento Art and Theory* (1947; reprint, Westport, Conn., 1971). For the concept of individual style, see Martin Kemp, " 'Equal excellences': Lomazzo and the Explanation of Individual Style in the Visual Arts," *Renaissance Studies* 1 (March 1987): 1–26. The importance of the Renaissance for the notion of originality in authorship is elaborated by David Quint, *Origin and Originality in Renaissance Literature: Versions of the Source* (New Haven, Conn., 1983). Further significant studies concerning the complex subjects of authorship, imitation, and originality in literature include Thomas M. Greene, *The Light in Troy: Imitation and Discovery in Renaissance Poetry* (New Haven, Conn., 1982); Deborah N. Losse, "From *Auctor* to *Auteur*: Authorization and Appropriation in the Renaissance," *Medievalia et Humanistica* 16 (1988): 153–63; and A. J. Minnis, *Medieval Theory of Authorship: Scholastic Literary Attitudes in the Later Middle Ages,* 2d ed. (London, 1987), esp. pp. 154–59. Many geographic areas and disciplines may have been affected by the notion of originality. For instance, Michael Baxandall, *The Limewood Sculptors of Renaissance Germany* (New Haven, Conn., 1980), pp. 123–27, discusses the new ideal of originality that developed among German mastersingers in the 1480s.

[95]Robert K. Merton, *Science, Technology and Society in Seventeenth-Century England* (1938; reprint, New York, 1970), p. 169, n. 30, "Priorities in Scientific Discovery: A Chapter in the Sociology of Science," *American Sociological Review* 22 (1957): 635–59, and "STS: Foreshadowings of an Evolving Research Program in the Sociology of Science," in *Puritanism and the Rise of Modern Science: The Merton Thesis,* ed. I. Bernard Cohen (New Brunswick, N.J., 1990), pp. 334–71, esp. pp. 343–48. See also Alan G. Gross, "The Rhetorical Invention of Scientific Invention: The Emergence and Transformation of a Social Norm," in *Rhetoric in the Human Sciences,* ed. Herbert W. Simons (London, 1989), pp. 89–107.

[96]The controversy that began in the 1580s between Tycho Brahe and Nicolaus Reimarus Ursus over priority in discovering the geoheliocentric system is only one of many examples. See Nicholas Jardine, *The Birth of History and Philosophy of Science* (Cambridge, 1984); Edward Rosen, *Three Imperial Mathematicians: Kepler Trapped between Tycho Brahe and Ursus* (New York, 1986); Owen Gingerich and Robert S. Westman, essay

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precisely, as well as the attitudes toward authorship and intellectual property that fueled them. Priority issues were intensified because new incentives (including patronage and the printing press) had developed for writing openly about discoveries and inventions. The ideal of openness as a value gained importance at the same time that patents and copyrights proliferated.[97] Issues of openness, secrecy, and the protection of intellectual property occasioned intense conflict in the 16th century. Although these conflicts need more study, it is clear that they were at the center of the development of early-modern scientific and technological thought and practice. Their investigation will illuminate early-modern science and technology as much as it will add historical understanding to present-day intellectual property issues.

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review of Rosen's work in *Isis* 79 (June 1988): 297–99, and *The Wittich Connection: Conflict and Priority in Late Sixteenth-Century Cosmology* (Philadelphia, 1988).

[97]See esp. William Bouwsma, "The Renaissance and the Broadening of Communication," in *Propaganda and Communication in World History*, vol. 2, *Emergence of Public Opinion in the West*, ed. Harold D. Lasswell, Daniel Lerner, and Hans Speier (Honolulu, 1980), pp. 3–40. For the ideal of openness in one particular tradition, see Pamela O. Long, "The Openness of Knowledge: An Ideal and Its Context in 16th-Century Writings on Mining and Metallurgy," *Technology and Culture* 32 (April 1991): 318–55.

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